

Equity Price Discovery with Informed Private Debt*

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Abstract — Equity markets fail to account for value-relevant non-public information enjoyed by syndicated loan participants and reflected in publicly-posted loan prices. A long-short strategy that buys (sells) the equities of firms whose loans have recently appreciated (depreciated) earns large risk-adjusted returns, suggesting a surprising and economically important level of segmentation across the same firm’s capital structure. The information lag captured by trading strategy returns is not affected by drivers of firm-specific attention, including publication of loan returns in the Wall Street Journal. Instead, returns to the strategy are eliminated among equities held by mutual funds that also trade in syndicated loans.

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JEL Classification: G11, G12, G14, G21, G23.

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1. Introduction

When prices are fully revealing, value-relevant information reflected in the price of one claim should be quickly integrated into the prices of related claims, even when market participants possess disjoint information sets (e.g., Grossman (1976)). In this paper, we use the secondary market for syndicated loans as a laboratory where participants in an increasingly liquid asset class receive privileged disclosures in advance of public markets. The setting allows us to evaluate the speed with which private information embedded in prices is recovered in related markets. We find that material non-public information reflected in publicly-available loan prices takes more than a month to appear in the price of the same firm's equity. The returns to an implementable trading strategy that takes advantage of the delay in information transmission suggest an economically significant failure of market integration within a given firm's capital structure.

Our examination of price discovery from the secondary loan market to public equities is motivated by the loan market's potential as a conduit for private information concerning firm value.¹ Unlike public bond markets, which share a common information environment with public equities and which Hotchkiss and Ronen (2002) show are tightly integrated with stocks, loan market participants such as banks are exempt from fair disclosure rules and enjoy a significant flow of material non-public information from borrowers. While loan investors may elect to be on the "public side" of a loan transaction (i.e., they agree not to receive non-public information) in order to trade in related securities, active monitors of the debt such as banks receive non-public disclosures in the ordinary course of a loan. This includes frequent disclosures detailing borrowers' often monthly financial statements, projections, covenant compliance certificates, amendment requests, acquisition and divestiture plans, and collateral account valuations (Gustafson, Ivanov, and Meisenzahl, 2016). Given the non-public nature of this information, private investors are restricted from sharing information directly or trading in the stocks of the same underlying firm.²

¹Meanwhile, the public equity market serves as a natural benchmark given its liquidity, the availability of traded prices, and the presumption of its relative efficiency.

²Despite this restriction, recent work suggests that institutional participants in loans may improperly

Yet critically, because syndicated loans are not considered securities, nothing prevents lenders from trading loans while in possession of private information and thereby communicating their information in the form of prices. In fact, as the liquidity of the loan market has grown, the demand for mark-to-market pricing has resulted in publicly posted daily dealer quotes mounting to roughly 3,700 names in 2014-2015 (the last two years of our sample). Meanwhile, over the normal course of business, debt and equity values co-move strongly in the same direction (Kwan, 1996), consistent with both loading predominantly on news about the value of the firm (we confirm the same across loan and equity markets).

Given private debt markets' informational advantage, the availability of public quotations, and the obvious implications for equity value, we might expect the timely integration of loan information into associated equity prices. To test this hypothesis, we form a zero-cost portfolio that buys the equities of recent winners in the loan market and sells recent losers. Under the null of fully revealing prices, this strategy should earn returns that are economically and statistically indistinguishable from zero. Instead, we show that, over a 17-year time period from 1998 to 2015, there is a one-month lag in the response of equity prices to the news embedded in loan prices.

While we go on to document the market dislocation in a variety of ways including panel and Fama-MacBeth regressions, the returns to the trading strategy uniquely highlight the economic magnitude of the market segmentation. Monthly abnormal returns are as large as 2.2% and support an annualized Sharpe ratio of 1.3. The observed degree of segmentation is consistent across segments of the equity market, including among the largest firms in the economy.³ For reference, the Sharpe Ratio of the well-documented

trade in public markets nonetheless (examples discussed below include Ivashina and Sun (2011), Massoud, Nandy, Saunders, and Song (2011), and Bushman, Smith, and Wittenberg-Moerman (2010)).

³Consider, for example, the case of Green Mountain Coffee Roasters (GMCR), later known as Keurig Green Mountain. In May of 2012, Green Mountain was a publicly traded company, with shares trading on the Nasdaq exchange under the GMCR ticker. In early May, GMCR closed on a syndicated loan facility, with the junior-most tranche beginning to trade at a Bid of 96 on May 6th. For the next 20 months, up to January 13, 2014, the quotes on this particular loan remained at 96. On January 14th, the bid on the loan jumped to 97.5, representing a 1.56% increase. In contrast, GMCR stock exhibited no strong movement on the same day, closing at \$80.35, up 1.18% from the previous day's close. Furthermore, January 14th featured no major public news about the company. About three weeks later, on February 5th, GMCR announced after the close of trading that Coca-Cola had purchased a 10% stake in the

momentum strategy over the same time period is 1.2.

The failure of equity markets to respond to the apparent profitability of following posted debt prices might immediately suggest a risk-based explanation. Yet we find that across several factor models that control for risks related to the market, size, value, momentum, short- and long-term reversals, liquidity, and distress, the economic and statistical significance of the strategy returns cannot be explained by factor exposures. As a companion to our portfolio sorts, we estimate Fama and MacBeth (1973) predictive regressions, where we control for firm characteristics known to predict excess stock returns that might be correlated with loan returns. As in the portfolio sorts, we find a strong predictive relationship between syndicated loan returns and subsequent excess stock returns. Furthermore, we find qualitatively identical results when we estimate panel predictive regressions with both fixed effects and double-clustered standard errors on the firm and time dimensions. Finally, we show that the strategy is not limited to stocks with binding arbitrage constraints.

Instead, we demonstrate that loan revaluations have stronger predictive power when we focus on high-yield debt, consistent with riskier debt inducing lenders to produce more non-public information (Gorton and Pennacchi, 1990). Furthermore, we see evidence consistent with loans anticipating news about earnings in advance of equities. In particular, we demonstrate that the predictive content of loan returns can be almost entirely explained by ex-post realizations of earnings surprises, analyst earnings forecasts, and credit rating changes announced over the ensuing twelve months.

The economic significance of returns to our simple trading strategy naturally lead us to examine what prevents equity market participants from fully integrating prices in the two markets. Said otherwise, what factors help or hinder the interpretation and incorporation of price signals into related markets? Investor inattention has been proposed by many as a central friction preventing the timely incorporation of public information in a general

company for \$1.25 billion, with the purpose of jointly developing and launching a new Green Mountain cold drink machine. The next day, GMCR stock opened at \$110.00, up 36% from the previous day's closing price of \$80.88. GMCR stock closed the month at \$109.78, up 35.5% from its beginning-of-month price. Importantly, the GMCR syndicated loan that had appreciated in mid-January had no subsequent price revisions, remaining steady at 97.5 through the end of February 2014.

sense (e.g., Hirshleifer, Lim, and Teoh (2011)), and could serve as a plausible friction inhibiting the integration of prices as well. Conditional on observing and allocating directed attention to loan prices, it seems reasonable that equity holders would extract the value relevance of changes in loan valuations and appropriately revalue equity claims. But if loan prices are not salient, or worse, investors are unaware of their availability, we might still expect loans to lead their associated stocks. We test this attention-based explanation by examining whether variation in the salience of loan- and firm-specific news can help to facilitate market integration.

We begin by exploiting the fact that, from 2000 to 2015, the Wall Street Journal (“WSJ”) reported once a week on the prices of syndicated loans, covering the top 25 biggest movers, along with dealer quotes for those names. We interpret this as a shock to both the availability and salience of loan market information and predict that, if inattention is segmenting markets, then reporting returns for a subset of names should reduce the profitability of our trading strategy. Instead, we find that over the course of our sample, a long-short portfolio buying WSJ-reported winners and selling WSJ-reported losers earns a monthly alpha of 2 to 2.5%. In other words, even when loan market information is presented prominently in a widely read financial periodical, equity market participants fail to incorporate that information in a timely fashion.

We also examine whether more general drivers of firm-specific attention can induce equity markets to more carefully incorporate price information. In particular, we consider the level of analyst coverage, changes in analyst forecasts, and periods with more firm news coverage. In sum, we find little support for the idea that attention directed to either the firm, or the firm’s loan prices, helps to facilitate the incorporation of loan return information in equity prices.

Our second hypothesis, and the one for which we find more support, is that extracting the information content implicit in loan prices requires expertise. Perhaps specialized equity investors are unable to interpret information embedded in debt prices, or discount the possibility that debt investors know something not already impounded in equity

prices.⁴ This specialization hypothesis would predict that market integration should be at least partially a function of portfolio integration. To the extent that debt and equity desks trade side by side and equity traders can enjoy some low-cost level of loan market expertise, the markets should move together closely.

We explore this idea by examining the effect of hybrid funds holding both loans and equities on the profitability of our strategy. Beginning in 2010, mutual fund holdings data began including information on fixed income and, in particular, syndicated loan holdings. After that point in time, we see a steady rise in the number of funds that own both equities and loans. We conjecture that such funds will better understand the value-relevance of loan prices and be able to take advantage of it by trading in the linked equities. Indeed, re-examining our portfolio strategy in this light, we find that stocks held by so-called integrated funds (those that hold both loans and equities) respond more quickly to price changes in the loan market. We argue that this suggests market integration is in large part driven by portfolio integration.

To the extent that the integration of loan and equity prices requires some form of expertise, we further demonstrate that this expertise appears to be fixed at the fund level. We find that cross-market expertise neither extends across funds within a fund family, nor does it appear to be a portable skill possessed by individual fund managers. In particular, we find no evidence that non-integrated funds led by managers with a history of overseeing integrated portfolios contribute to more tightly linked loan and equity markets.

Our paper extends a well-established literature documenting the integration of price signals across markets and across the firm's capital structure. For example, Gagnon and Karolyi (2010) show that discrepancies in cross-listed equity prices are economically small. Easley, O'Hara, and Srinivas (1998) demonstrate that signed volume in the option market leads stock prices at just five-minute intervals. Most closely related to our paper

⁴For example, in the model developed in Vives and Yang (2017)), agents' sophistication dictates their ability to comprehend the price function and thus recover information implicit in prices. They show this limited sophistication can equivalently be interpreted as an investor's choice to not pay attention to some prices in the economy.

is that of Hotchkiss and Ronen (2002), who examine lead-lag effects between corporate bonds and equities at the intraday frequency and find that prices in both markets respond equally quickly to earnings news.

Our broad reading of these studies suggests that modern markets in the United States are generally well-integrated. Yet the mechanism by which market integration occurs in practice is not always clear. On one hand, we often assume that participants in markets with disjoint information sets learn from each other's prices (e.g., Grossman (1976), Radner (1979), Grossman and Stiglitz (1980), Hellwig (1980)). On the other hand, this is difficult to empirically distinguish from prices moving together as participants respond to common information in parallel. In contrast to the studies above, which compare public securities markets where investors share a common set of public information about the firm, our focus on the relationship between private and public claims enjoying disparate levels of disclosure is novel and creates an important wedge in the information environment across assets. This allows us to ask about public investors' ability to recover information from the price signals created by participants receiving privileged disclosures.

We also build on earlier work documenting the pricing properties of syndicated loans, which make up a critical part of the typical U.S. firm's capital structure and have evolved into an increasingly liquid asset class; syndicated loans make up as much as 50% of all corporate financing (Sufi (2007)) and more than 40% of syndicated loans were traded as of 2013 (Beyhaghi and Ehsani (2017)). Particularly relevant are earlier papers documenting the informational advantage of private debt over equity, including Gande, Altman, and Saunders (2006, 2010), who examine the price anticipation of ex-post default events and find that loan market prices reflect these events well in advance of equity markets. Allen and Gottesman (2006) also examine the lead-lag relationship between loan and equity returns. Using data from 1999 to 2003, they show that weekly loan returns Granger cause future equity returns, but find that a trading strategy based on loan market returns fails to reject cross-market integration.⁵ Ivashina and Sun (2011) and Massoud, Nandy, Saunders,

⁵Their failure to reject cross-market integration is driven by both a shorter time series and only looking at weekly return lags. In Appendix A, we show that equity markets gradually incorporate information anticipated in weekly loan returns over horizons that exceed one week.

and Song (2011) show that institutions appear to engage in insider trading related to the private information generated by lending relationships and correspondingly earn excess returns. Bushman, Smith, and Wittenberg-Moerman (2010) suggest that, as a result, equities benefit from faster price discovery around earnings announcements when firms' lenders receive early information via covenants or other forms of monitoring.

Our findings are consistent with private lenders possessing and perhaps even trading on private information but suggest that, to the extent information leakage does occur, it is insufficient to integrate markets. The remaining predictability translates into a large and meaningful economic magnitude when presented as the return to a trading strategy. This result is especially surprising in light of the fact that price quotes in the active secondary market for private debt claims are publicly available. Hence, no insider trading or direct disclosure of private information should be required to fully integrate private lender information into other markets.

Finally, our results are of potential importance for policymakers. Given lenders' access to material non-public information about their borrowers, how that information is protected when the secondary market for loans becomes a price discovery market is an open question. In particular, there is an apparent disconnect between SEC mandates for lenders to keep private information private and the failure to prevent the efficient transmission of information through dealer quotes. A solution that precludes shutting down liquidity in the secondary market or limiting bankers' access to non-public information, both of which would have significant consequences for credit markets, is difficult to imagine. In the meantime, our findings that stocks held by institutions with fluency in both markets are better integrated suggest that some firms are able to take advantage of this privileged information, lending support to theories of information disclosure (e.g., Vives (2010), Goldstein and Yang (2017)) in which public signals of fundamental value improve market efficiency. Importantly however, we provide evidence consistent with the idea that when investors' capital is segmented across markets (e.g., Duffie (2010)), price efficiency can be slow, or even fail, to materialize (e.g., Goldstein, Li, and Yang (2014)).

2. Background and Data

2.1. Loan markets

Our choice of the secondary loan market as a laboratory for understanding information transmission is motivated by the fact that loan prices may reflect material non-public information for reasons described below. However, the importance of the syndicated loan market suggests that a better understanding of associated price discovery is also of interest more generally. In particular, Sufi (2007) points out that syndicated loans are a cornerstone of large firms' capital structure, representing 51% of U.S. corporate finance originations. Further, syndicated loans support more underwriting revenue than both the public equity and bond markets. Although the motivations for issuing in the loan market are as diverse as the firms that do so, in contrast to public debt, loans can be more flexible in drafting the terms of the credit, tend to be easier to renegotiate, and provide borrowers with diversification in their access to funding markets.

Meanwhile, these loans are increasingly traded in the secondary market. In 2015, of the top 500 firms in the CRSP universe, 30% had syndicated loans that were quoted in the secondary market. Further, this already large market continues to grow. In the last two years of our sample, 3,724 firms had daily loan quotes, compared to 1,482 firms in the NYSE and 4,223 in the total CRSP universe over the same time period.⁶

The secondary loan market emerged in the 1990s out of banks' desire to trade out of primarily distressed positions. However, as non-bank lenders including CLOs, insurance companies, mutual funds, and hedge funds became increasingly active loan investors, the need for broad-based liquidity motivated the formation of the Loan Syndications and Trading Association (LSTA) in 1995. Tasked with the development of the secondary market, the LSTA's efforts included the establishment of a mark-to-market pricing service based on dealer quotes in 1999. These same dealer quotes form the basis of our loan price

⁶In theory, firms can limit the trading of their loans in secondary markets. While some do, this is increasingly rare. Drucker and Puri (2009) show that traded loans facilitate more durable lending relationships and more credit availability. Pyles (2008) shows that allowing loans to be traded affords borrowers lower ex-ante costs of credit.

signal, discussed in detail in the next section.

The trading of syndicated loans provides a unique setting in that it allows us to observe publicly available daily prices for private claims – that is, claims that are not public securities under U.S. securities law and hence can be traded by institutions in possession of non-public information (Taylor and Sansone, 2006). Furthermore, banks and non-bank lenders are provided carve-outs under fair disclosure rules in order to receive non-public information in the course of their monitoring. This may include monthly financial statements, collateral valuations (e.g., receivables), notification of covenant violations or amendment requests, and participation in early conversations about acquisitions or asset sales that might violate loan terms. In addition to loan-specific information provisions, standard credit agreements allow for catch-all provisions that require borrowers to provide timely information “regarding the operations, business affairs, or financial condition” that any lender may reasonably request (Wight, 2009).

In combination, the facts above suggest that: i) lenders have access to a regular stream of privileged information about the borrower, ii) lenders are allowed to trade their loans with each other or others while in possession of that information, and iii) the prices at which lenders with privileged information are willing to trade are publicly reported. As a result, it is commonly suggested that material non-public information available to lenders but not to public markets may be telegraphed into loan prices in advance of public disclosures. For example, Standard and Poor’s writes in its U.S. Loan Market Guide that “mark-to-market pricing services... from time to time report significant movement in a loan price without any corresponding news. This is usually an indication that the banks have received negative or positive information that is not yet public.” This informational advantage of private lenders allows us to then ask how adept public equity markets are at recovering information from prices when some debt investors enjoy privileged disclosures.

2.2. Loan data

Our tests of the equity market’s response to news implied by private debt prices require a matched dataset of loan and equity returns. The loan data come from Thomson Reuters

and the Loan Syndications and Trading Association, which collect and aggregate dealer quotes for widely traded syndicated loans. Since 1999, prices have been quoted daily and distributed with a morning-to-afternoon lag. Dealers submit indicative prices to the data provider in the morning, those prices are audited over the course of the afternoon, and cleaned prices are then distributed by 4pm (via a website, or in the past, a secured internet delivery system). These prices are used widely as a source of mark-to-market pricing for loan market investors, including both banks and non-bank institutions.

Note that the dealer quotes are only quotes and do not reflect actual transaction prices. Moreover, while they are described by the provider as quotes at which the dealers would be willing to buy or sell, there is little guidance as to the size of trade that one could actually execute at the reported bid or ask. Importantly, most of the quotes we observe are averages of individual quotes submitted by multiple dealers. Thus, even in months with non-zero price movements, there may be some dealers who do not actually update their quotes.

In short, there are reasons to be concerned that the quotes are both stale and perhaps not reflective of prices at which one could actually trade. Hence, while it is tempting to wonder about the extent to which one could trade profitably in the loan market on public information, our data are not likely to shed conclusive light on that question.⁷ Instead, we rely on the loan quotes as a signal on which to trade in other more liquid markets for which transaction data are available. Based on its relative liquidity and presumed efficiency, we choose the equity market as a natural benchmark. Because of the risk of latency in loan quotes, we also restrict our equity trading to the monthly frequency based on monthly loan signals. At any higher frequency, we observe very little movement for a typical loan.

The median loan in our merged sample has daily quotes from two dealers (average of 2.75), typically large banks, although depth grows over time within the sample. At a minimum, the lead arranger/administrative agent for the loan at origination will remain a

⁷Using loans with single dealer quotes, a subsample where we can be sure that quotes are not stale due to a subset of dealers not updating their quotes, we show that public equity returns fail to predict loan price movements. See Appendix B for details.

dealer in the secondary market for these loans. We include all U.S. dollar currency loans, including term loans, both so-called A and B tranches (or TLA and TLB), respectively designed to be held by banks and non-banks, as well as revolvers, typically held only by banks. Roughly a quarter of the loans in the sample are revolvers. 30% are designated TLB and 21% are designated TLA or simply term loans.

These are floating-rate loans, with an average spread of 273 bps over LIBOR. They also trade at discounts, with an average bid of 95.9 and an average ask price of 97. The loans have a median maturity of six years, although the average loan appears in the mark-to-market database for only 23 months (from first appearance to last). The average borrower will have several loans over the course of the sample, some of which may overlap. The median (mean) borrower has five (7.25) distinct loans traded.

Although we have referred above to “loan returns” as a potential trading signal, because spreads on the loans are unaffected by new information received by lenders, we focus our attention instead on the price appreciation or depreciation that occurs for a given loan over a given month to track new information acquired by private lenders. Meanwhile, because of the likelihood of stale pricing discussed above, in many cases, we ignore loans for which prices did not move in a given month. In the common event that a borrower has multiple loans outstanding in a given month, we focus our attention on the price movement of the cheapest loan – that is, the loan with the highest effective spread (the spread over LIBOR offered in the contract, plus any capital gain or loss a lender holding the loan to maturity would earn assuming repayment). By focusing on the riskiest debt claims, we capture more variation in pricing signals, as well as variation that is more likely to be relevant to equityholders. Finally, we use the midpoint between the average bid and the average ask price as the relevant measure of price and calculate returns as the percentage change in price.

2.3. Matching stock and loan data

We obtain monthly stock returns, stock prices, and shares outstanding from the Center for Research on Security Prices (CRSP). We limit our analysis to common shares, those

with share codes of 10 or 11. To eliminate concerns related to illiquidity among small stocks, we include only stocks with market capitalization above the 10th percentile of NYSE breakpoints at the time of portfolio formation. Further, we restrict the sample to include only firms with a nominal share price of at least \$1 at portfolio formation. Given a monthly loan return for a specific borrower, we match borrowers to their traded stocks using the Dealscan–Compustat links produced by Michael Roberts and Sudheer Chava as of 2012 (Chava and Roberts, 2008) and extended through 2015, along with the CRSP–Compustat linking tables.⁸

2.4. Other data sources

We also obtain monthly Fama-French factor returns over the same period from Ken French’s online data library. Monthly data on the liquidity factor (LIQ) are obtained from Lubos Pastor’s website, and monthly betting against beta factor (BAB) returns for U.S. stocks are downloaded from AQR’s online data repository. We also construct a factor for distress following Campbell, Hilscher, and Szilagi (2008) using a combination of Compustat and CRSP.

We use data from Compustat to calculate book-to-market ratios for each public firm in our sample. The book-to-market ratio is defined as year-end book equity plus balance sheet deferred taxes scaled by the year-end market value of equity. This calculation is implemented after imposing the usual six-month lag to ensure the observability of measured values.

Finally, we use data on quarterly mutual fund holdings from the CRSP Survivor-Bias-Free U.S. Mutual Fund Database. We focus on funds that hold both stocks and syndicated loans. Stocks held by U.S. mutual funds are identified by *permno*. To identify syndicated loan holdings, we implement a partial string matching algorithm that searches for security names that include the strings *synd*, *loans*, or *lns*. We then inspect all matches by hand to verify the accuracy of the algorithm. Because of data limitations, our final

⁸The match between traded loans and Dealscan is provided by Thomson Reuters, which owns both databases. With few exceptions, traded loans covered by the LSTA data are a subset of loans covered by Dealscan.

sample of mutual fund holdings spans the period from September 2010 to June 2015.

2.5. Summary statistics

We end up with 18,335 monthly matches of loan returns and linked equity returns covering 920 unique firms over the period from September 1998 to August 2015.⁹ Over the course of this sample, we always have a minimum of 30 matched stocks in a given month. Table 1, Panel A presents summary statistics for the sample of matched loan and stock returns. The mean and standard deviation of loan returns in the sample are -0.078% and 2.480% , respectively. For the same firms, excess stock returns in the next month have a mean of 0.674% and a standard deviation of 15.647% .

For purpose of comparison, we also report statistics for the overall CRSP sample during the same sample period. The size statistics indicate that firms with traded loans are economically significant; the average firm in the sample has a market capitalization of about \$1.63 billion, almost five times larger than the average CRSP firm over the same sample period. Larger size affords these firms more liquidity in equity markets, with significantly lower bid-ask spreads. Hence, although we will decompose our results based on liquidity later, we begin with a relatively liquid sample of firms. Where firms in our sample are distinct is in their use of leverage. In particular, the average market leverage of firms with traded loans is 1.27, significantly higher than the representative CRSP firm (at 1.08). Meanwhile, the median issuer credit rating in our sample is BB- (compared to BBB- among CRSP firms). As a result, when we consider risk associated with these firms, we will need to be careful to consider distress as a relevant factor.

It is also important to note that contemporaneous loan returns and stock returns are strongly positively correlated. In pooled regressions with time (and firm) fixed effects, stock returns load on loan returns with a coefficient of 0.75 (0.73) and a t -statistic, calculated using standard errors clustered by firm and time, of 6.27 (6.17). In Fama-MacBeth regressions, the cross-sectional loading is 1.3 with a t -statistic of 6.7 (calculated

⁹Removing the restriction that loan returns are non-zero increases the sample to 34,394 observations covering 1,014 unique firms, a sample where we verify that our baseline results are robust.

using Newey and West (1987) adjusted standard errors). These results are consistent with those of Kwan (1996), who shows a positive relationship between bond and equity returns, and confirms that, on average, good news for loans is good news for equities and vice-versa. In other words, while we cannot rule out that on occasion, risk-shifting may drive the value of claims in opposing directions, this would seem to be the exception and not the rule, and would work against our finding a result.

3. Market Integration Tests

Our analysis is based on the conjecture that publicly observable prices in the syndicated loan market incorporate private information available to dealers. The predictive power of loan returns over next month's stock returns provide a joint test of 1) whether or not lenders are informed, and 2) whether or not equity investors have the capacity to recover lenders' information from prices in a timely fashion. Formal cross-sectional tests in the next section are motivated by Figure 1, which plots pooled variation in loan returns against equity returns for the same set of borrowers over the subsequent month. The x -axis sorts borrowers into 25 quantiles based on mean monthly loan return, ranging from -9.33% to 8.06% in the extremes. For the same set of borrowers, the y -axis shows the average equity returns in the subsequent month for each bucket of firms and finds a spread of almost 10% among the most extreme loan market winners and losers.¹⁰

As we demonstrate in the following sections, similar variation can be captured using dollar-neutral portfolios constructed in real time using cross-sectional sorts of equities based on publicly available loan performance information. Importantly, unlike statistical tests of Granger causality designed to detect the presence of lead-lag relationships across markets, using the returns to a tradable portfolio that takes advantage of loan prices in real time allows us to recover both the statistical *and* the economic significance of any

¹⁰Although the relationship between loan and subsequent equity returns in Figure 1 appears to be driven by the extremes, it is important to note that loan returns do not move linearly across quantiles. In untabulated tests, we find that extreme loan returns exhibit, if anything, a flatter relationship than loan and equity returns in the middle buckets. Consistent with this observation, we verify that our baseline market integration tests are robust to dropping the top and bottom 10% of loan returns each month.

observed segmentation. Of course, one challenge to interpreting the returns from portfolio sorts is the multitude of potential risk-based stories. To help separate the two related explanations, we utilize standard methodologies in the empirical asset pricing literature.

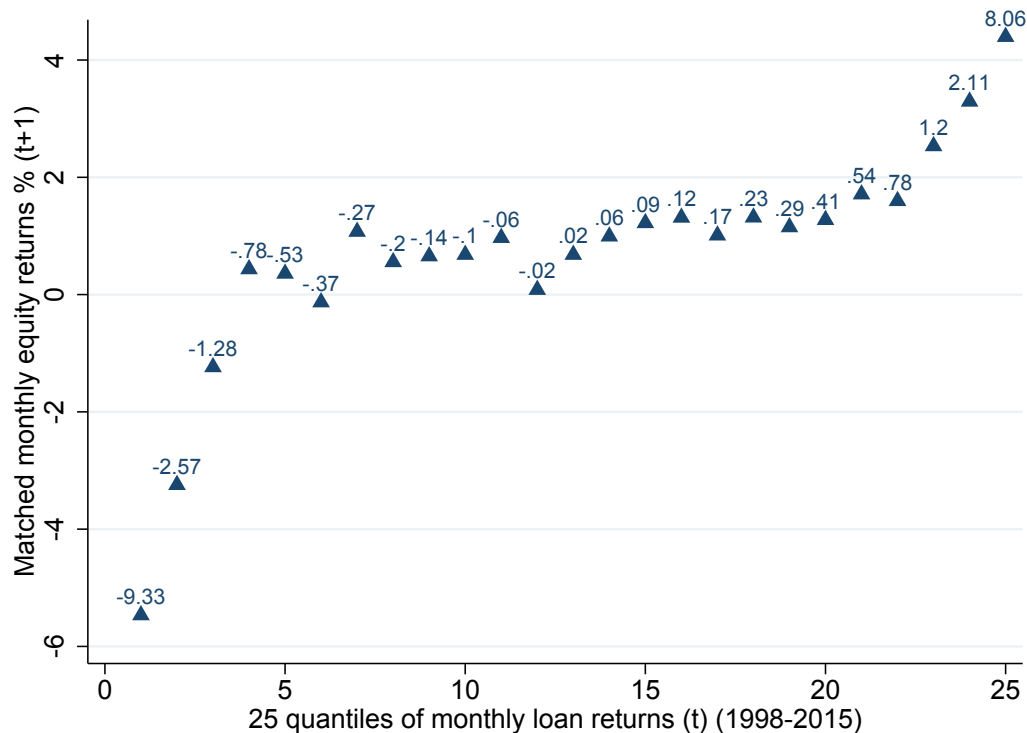


Figure 1

Equity Returns vs. Lagged Loan Returns. The plot above combines both time-series and cross-sectional relationships between loan returns and linked equity returns in the next month. Loan returns are sorted into 25 quantiles over the entire sample period and plotted against the next-month equity returns for the same borrowers. Average loan returns for each quantile are reported next to the triangles.

3.1. Construction of stock portfolios

In our first test of integration between the secondary loan and equity markets, we perform univariate sorts. Specifically, we sort all stocks with a matched non-zero loan return in month t into quintiles.¹¹ We then form six portfolios and track the performance of each over month $t + 1$.

The short portfolio contains the quintile of stocks with the lowest observed loan returns in month t . The long portfolio contains the quintile of stocks with the highest observed

¹¹As a robustness check, we verify that our results hold when we include zero loan returns in the sorts. We also verify that our results hold when we alternatively sort into terciles and deciles.

loan returns in month t . We then form the long-short portfolio, a dollar-neutral portfolio that captures the difference in returns of the long and short portfolios in month $t + 1$. Finally, portfolios 2 through 4 contain the remaining quintiles of stocks sorted on loan returns in month t .

On average, each of the 920 firms in the sample appears in the one of the portfolios for a total of about 20 months. Further, the average firm has 5.6 distinct appearances in the portfolios, with the mean appearance lasting 3.6 months. Overall, the portfolios exhibit relatively frequent rebalancing, with monthly average turnover of 81%. In over 98% of cases, a firm exits the set of portfolios because its loan return is equal to zero, suggesting that lenders' private information flows are generally uninformative, with the exception of extreme cases. For 91 cases in our sample, firms permanently exit because they are delisted (there are no temporary delistings where firms re-appear in our sample). Of these delistings, only 7 are related to poor performance (e.g., price too low, bankruptcy, and filing delinquencies), with the remaining 84 being the result of mergers.¹² For 81 of the 91 total delistings in the sample, we use CRSP delisting returns. For the remaining 10 delistings where CRSP's delisting return is missing, we use the last available full-month return in CRSP.¹³

3.2. Portfolio sort results

The portfolio performance estimates are presented in Table 2, where we report raw equal-weighted returns for each portfolio. We also report CAPM alphas, as well as alphas from 3, 6, and 9-factor models. The 3-factor model includes the excess market return (RMRF), the value factor (HML), and the size factor (SMB). The 6-factor model adds the momentum factor (UMD) as well as the short- and long-term reversal factors (STR and LTR). These standard benchmark models help control for exposure to risk and ensure

¹²Note that the proportion of delistings in our sample compares favorably to that in the CRSP universe. In particular, $91/18,335 = 0.50\%$ of our observations correspond to delistings vs. $7,696/948,573 = 0.81\%$ in CRSP during our sample period.

¹³Since this may impart a bias on returns to the trading strategy (Shumway, 1997; Shumway and Warther, 1999), we verify that our results are robust to using Shumway's (Shumway and Warther's) suggestion of replacing these returns with -30% (-55%). We also verify that our results are robust to dropping these 10 delisting observations, as well as to not using CRSP delisting returns at all.

we aren't recapturing known sources of excess returns.

In the rightmost column, we add additional factors for liquidity (LIQ), betting against beta (BAB), and distress. These risk factors are motivated by the setting of our study. For example, we note in the summary statistics presented in Table 1 that the long and short portfolios differ in terms of liquidity as measured by bid-ask spread. Meanwhile, the short portfolio has a higher CAPM beta than the long-side – this is evident from the increased alpha moving from raw-returns to a CAPM model, and can be confirmed in Appendix Table A1.¹⁴ This motivates the inclusion of the BAB factor. Finally, although the long and short portfolios do not appear different with respect to levels of creditworthiness (both have similar ratings distributions in Table 1, Panel B), changing loan prices are clearly related to changes in firm distress. As a result, we control for a monthly distress factor following Campbell, Hilscher, and Szilagi (2008). The *t*-statistics reported in parentheses below the coefficient estimates are calculated using Newey and West (1987) adjusted standard errors using a six-month lag.¹⁵

Across each model, we find evidence consistent with the loan market leading equities. The estimates in column 1 of Table 2 indicate that portfolio returns increase monotonically with syndicated loan returns. However, echoing our earlier pooled findings in Figure 1, the spread across the three middle portfolios is modest, with the bulk of the predictive effect residing in the extremes. In particular, the short portfolio generates an average monthly return of -0.535% , while stocks in the long portfolio earn 1.580% on average. The difference between the long and short portfolios amounts to an average monthly return of 2.115% . This monthly difference is highly statistically significant, with a *t*-statistic of 4.63.

The factor model alpha estimates presented in the remaining columns suggest that

¹⁴For brevity, we report only factor model alphas in Table 2. We present the full set of corresponding factor loadings for the long, short, and long-short portfolios in Tables A1 and A2 of the appendix.

¹⁵Standard practice for lag selection varies between choosing the smallest integer that exceeds $T^{1/4}$ and $T^{1/3}$, where T is the length of the time series (e.g., Greene (2007, pp. 463, 643), Newey and West (1994)). For our sample length (204 months), this translates to lags between 4 and 6 months. Thus, we consider lag lengths of 4, 5, and 6 months when calculating Newey and West (1987) adjusted standard errors and find that lag lengths of 6 uniformly generate the most conservative (i.e., smallest) *t*-statistics in our tests. Therefore, we report all results using the 6-month lag.

the economic and statistical significance of the long-short portfolio returns cannot be explained by factor exposures. Specifically, the long-short alphas range between 2.046 and 2.253% per month, and remain highly statistically significant (t -statistics between 4.43 and 4.89). Further, both the long and short portfolios contribute to the profitability of the dollar-neutral strategy. Figure 2 plots the cumulative abnormal returns from column 2 of Table 2. Although the profitability of the strategy appears to flatten in the latter half of the sample, splitting the sample confirms that even in the second half of the sample, the strategy generates an economically and statistically significant 9-factor monthly alpha in excess of 1%.

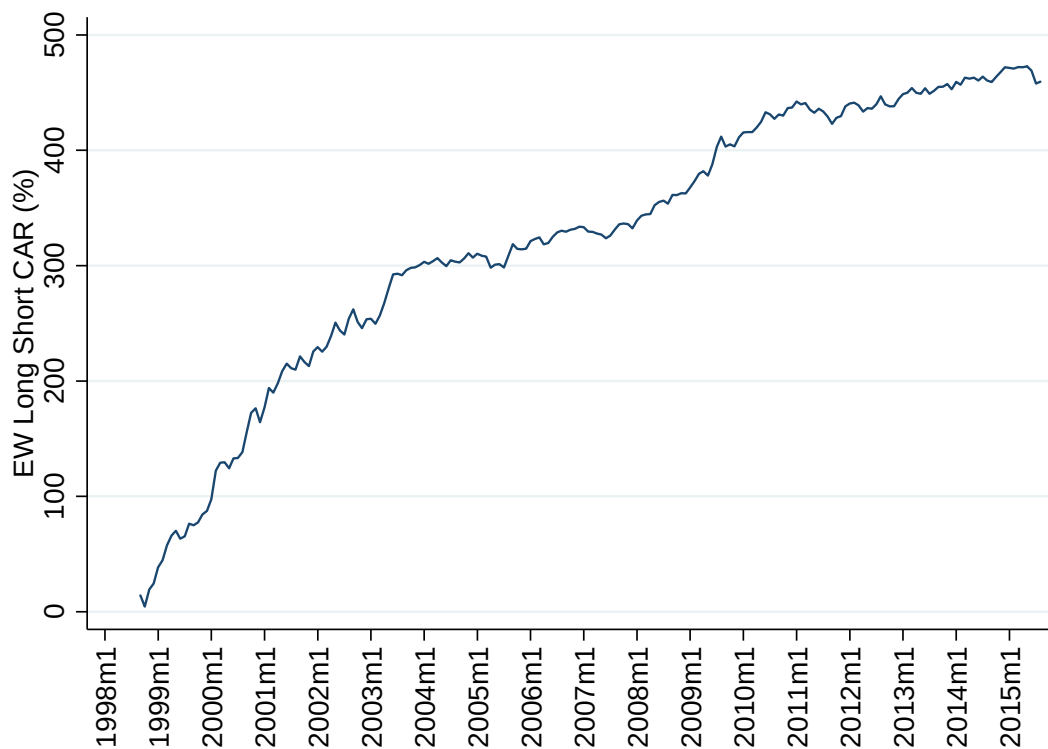


Figure 2

Cumulative Equal Weighted Portfolio Returns. We plot the monthly cumulative abnormal returns from the equal weighted long-short strategy, calculated as alpha plus the cumulative residuals from a CAPM regression over the sample period.

If the observed predictability is truly a result of informed participants in the loan market, then we would expect its effects to be strongest both where incentives to produce private information are highest, and in periods when lenders are mostly likely to have an informational advantage. To assess the first statement, the second panel of Table

2 exploits cross-sectional variation in how much private information loan returns are likely to contain based on ex-ante loan risk. Our conjecture is that riskier debt induces lenders to produce more non-public information (Gorton and Pennacchi, 1990), which in turn should show up in dealer quotes. To test this, we form double-sorted portfolios by first sorting on the effective spread of the loans at issuance and then on monthly loan returns. Although both the relatively safe and risky portfolios earn monthly alphas exceeding 90 bps, the long-short portfolio formed using the high spread loans generates a significantly larger excess return than that using relatively safer debt. While by no means conclusive, these findings are consistent with the stronger incentives to produce non-public information on risky debt, combined with equity investors who are slow to react to loan prices.

Meanwhile, regarding the timing of lenders' informational advantage, Figure 3 confirms that returns are strongest in months when the gap between public and private information about earnings is largest. The figure plots average returns to the equal weighted long-short strategy for each calendar month of the year based on the equity holding period. Dashed vertical lines highlight common annual and quarterly earnings announcement months. We note immediately that the predictability is highest in February, when the modal firm publicly announces annual earnings, and one month after privately informed lenders (who often receive monthly unaudited statements) would learn about earnings news. In fact, February returns are negative in only one month during the sample period. Further, we find that strategy returns exhibit a marked downtick in the month after earnings announcements, when information asymmetry between the loan and equity markets regarding earnings is likely to be lowest. Later, in Table 4, we decompose the information advantage of lenders behind the price lags and show that between the release of upcoming earnings information, revisions in analyst forecasts, and ratings changes, we can almost completely explain the leading content of loan prices. With quarterly earnings releases, the average firm is less than 2 months away from large public disclosure opportunities that serve to align private and public information sets. This puts a cap on how long the delay in market integration can survive – one that is

largely consistent with our roughly one to two month delay.

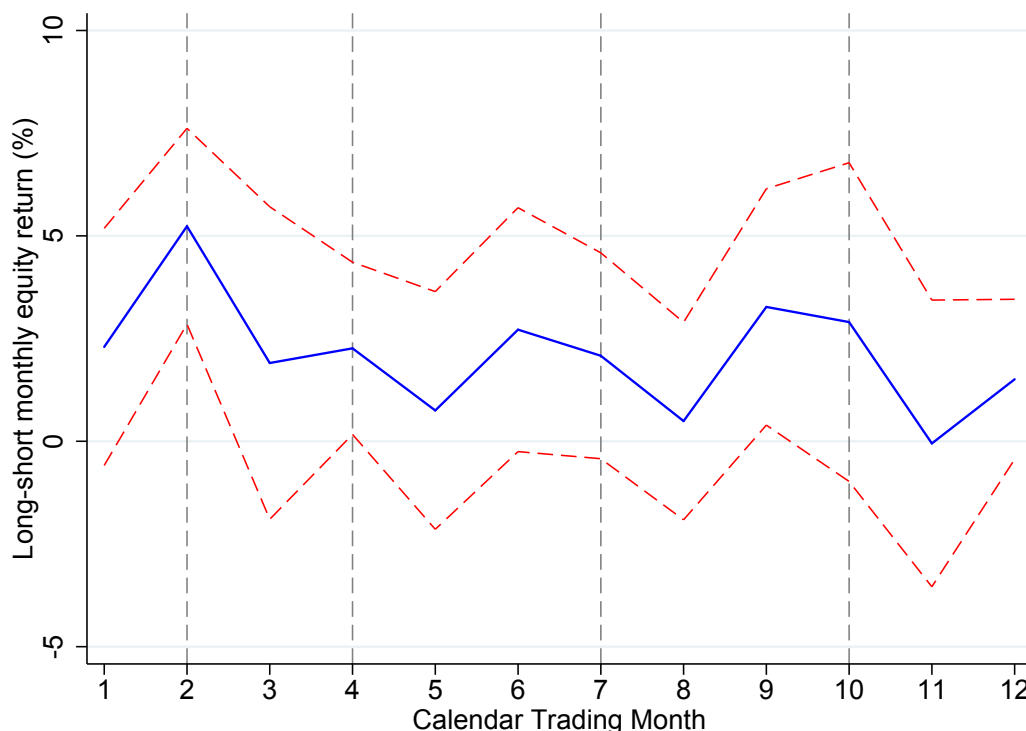


Figure 3

Calendar Month Returns. The solid line shows the average returns to the equal weighted long-short strategy grouped by calendar month, based on the equity holding period. Average returns are surrounded by ± 2 standard error bands (dashed lines). Dashed vertical lines highlight common annual and quarterly earnings announcement months.

Finally, our findings are only slightly attenuated when looking at value-weighted portfolio returns, with the average monthly raw long-short portfolio returning 1.356% (t -statistic = 2.78). Although the smaller magnitude of the value-weighted portfolio returns suggests that the predictive relation between loan and equity returns is stronger among smaller stocks, our main predictability results are not isolated to the smallest stocks. In Table A4 of the appendix, we rerun our analysis focusing only on the subset of stocks with market capitalization above the median NYSE size breakpoints and continue to find economically and statistically significant long-short portfolio returns among the largest stocks.¹⁶

¹⁶We also consider several other robustness checks in the Appendix. In particular, in Table A4, we show that we can limit the loan signals to those emerging from revolving loans, term loan Bs, and non-distressed loans. We also consider the robustness of the strategy to focusing on non-earnings announcement months and calculate turnover-weighted portfolio returns. Finally, in Tables A5 and A6,

In terms of economic significance, it bears mentioning that the portfolio is implicitly levered, by construction. This is due to the fact that all portfolio stocks have a significant volume of traded debt and that portfolio alphas and betas both scale with firm leverage. As a useful exercise to put our returns in perspective, we can de-lever the equal-weighted long-short strategy so that the average market beta of each side (currently 1.65 on the short side and 1.35 on the long side of the equal-weighted book) is equal to one. When we do that, the equal-weighted alphas are more consistent with prior results from the return predictability literature, at 1.5% per month. A similar exercise for the value-weighted strategy generates monthly alphas of 1.12% per month.¹⁷ Alternatively, it is perhaps easiest to interpret the strategies in terms of Sharpe ratios, which are on par with momentum, at 0.6 for the value-weighted strategy and 1.2 for the equal-weighted strategy. Thus, while nearly any information signal can generate statistically significant abnormal returns given a large enough sample and small frictions in capital markets, the sizable economic returns to incorporating obviously-relevant loan price information suggest large and important distortions segmenting the markets for debt and equity.

3.3. Fama-MacBeth predictive regressions and robustness tests

As a companion to our portfolio sorts, we estimate Fama and MacBeth (1973) predictive regressions, allowing us to control for a host of characteristics known to generate excess stock returns that might plausibly be correlated with loan returns. For each sample month, we regress excess stock returns in month $t + 1$ on a set of return predictors observable at the end of month t . Our main predictor of interest is each firm's syndicated loan return in month t .¹⁸

Of the firm characteristics known to predict excess stock returns, we begin by including

we examine the three way relationship between loans, bonds, and equities for the limited subsample where all three are traded. In this subsample, we rule out the possibility that the predictive content in loan returns observed in Table 2 is also reflected in bond returns. See Appendix C for details.

¹⁷We de-lever the long and short portfolios using the same factor in order to ensure that t -statistics on the alpha are unaffected. De-levering each side separately generates economically similar effects.

¹⁸We also implement our main regression specifications from Table 3 as panel regressions with both firm and time fixed effects, and standard errors clustered by firm and time. The results from these panel specifications are qualitatively similar to those in our main Fama-MacBeth regressions (see Table A7).

size and the book-to-market ratio. Size is calculated as the log of market capitalization and book-to-market is computed using information available at least six months prior to the end of month t . We also include the return over the previous six months, with a one-month lag, to capture momentum effects (Jegadeesh and Titman, 1993) and the contemporaneous stock return in month t to account for short-term reversals (Jegadeesh, 1990). The ratio of debt to equity accounts for the market leverage effect of Bhandari (1988). To capture news related to financial distress, we also control for expected default frequency based on S&P credit ratings as well as a credit rating up/downgrade indicator (± 1 for up/downgrades in month t).¹⁹ Finally, we control for standardized unexpected earnings (SUE) announced in months $t - 2$ through t (Livnat and Mendenhall, 2006). Leverage, expected default, and credit rating controls are natural given the sample, and SUE is inspired by the suggestion in Figure 3 that loans may contain earnings information.

We report the time series averages of monthly cross-sectional predictive regressions, along with t -statistics based on these coefficients, in Table 3. The t -statistics reported in parentheses below the coefficient estimates are calculated using Newey and West (1987) adjusted standard errors using a six-month lag.

As in the portfolio sorts, the estimates in Table 3 indicate a strong predictive relationship between syndicated loan returns and subsequent excess stock returns, even after accounting for a host of potentially correlated omitted firm characteristics. Stepping through different combinations of controls in columns 1-11, we find the coefficient on loan returns ranges from 0.387 to 0.570 (with t -statistics of 2.14 to 2.84). In economic terms, as a lower bound, our coefficients imply that a one standard deviation change in loan returns translates to a $0.387 \times 2.477 = 0.96\%$ increase in next-month excess stock returns after accounting for firm characteristics. Finally, in column 12, we include all controls simultaneously and interact the syndicated loan return with the size characteristic.²⁰ Echoing the difference in returns found in Table 2 where we compared equal- vs.

¹⁹Expected default frequencies for each rating category are taken at the five year horizon from S&P's 2016 annual global corporate default study. Using expected default frequencies avoids using an ad hoc numerical control for ratings. Moreover, we cannot use ratings fixed effects as non-parametric controls because many ratings may not be represented in any given cross-section for the Fama-Macbeth regressions, prohibiting consistent estimation of their coefficients.

²⁰We demean size in each cross-section before computing the interaction so that the syndicated loan

value-weighted portfolios, the significant negative coefficient on the size interaction indicates that the predictive power of the syndicated loan return is dampened for the largest firms in the sample, motivating our subsequent hypotheses related to both inattention and limits to arbitrage. However, the economically and statistically significant coefficient on the syndicated loan return predictor (coefficient = 0.522; t -statistic = 2.81) signals the existence of a significant predictive relation between syndicated loan market returns and subsequent stock returns for firms of average size.

In Appendix D, we explore several alternative interpretations of the relationship between loans and subsequent equity returns; notably a risk-based story and limits to arbitrage. Under the risk interpretation, changes in loan value might be associated with changing exposure to unobserved (and hence uncontrolled-for) risk factors. Under limits to arbitrage, predictability might be concentrated in shares of hard-to-trade stocks. In either case, higher equity returns in the month after elevated loan returns would not imply a failure of equity market participants to recognize the value-relevance of private debt prices. However, neither of these explanations is strongly supported by the data.

With respect to risk-based explanations, if loan returns signal changes in exposure to unobserved risk factors, then holding component stocks fixed, we would expect the differential between long and short portfolio returns to be relatively persistent²¹. Instead, Figure A2 in the Appendix shows that risk-adjusted returns to the dollar-neutral portfolio are earned primarily in the first month after the associated loan returns. This pattern appears to be more consistent with mispricing that is eventually corrected than a change in exposure to systematic risks among portfolio component stocks.

We also find little support for the idea that arbitrage constraints generate the delayed response to loan prices. In Table A8, we include interactions between several standard characteristics associated with limits to arbitrage (idiosyncratic volatility, institutional ownership, the bid-ask spread, and illiquidity) and the loan return predictor and find

return coefficient measures the predictive effect for a firm of average size.

²¹For example, consider a borrower who agrees to increase the spread on their loan to secure a covenant waiver or amendment. The higher spread would translate to higher financial leverage and greater exposure to systematic risks on the part of shareholders. Thus, expected and average realized stock returns would be persistently higher going forward.

that the interactions between the arbitrage constraint proxies and the loan predictor are indistinguishable from zero at standard levels of significance. We also consider to what extent the cost of shorting shares of recent losers in the loan market serves as a barrier to implementing the strategy, given that the value-weighted strategy earns much of its returns from the short leg of the portfolio. However, the dynamics of short interest among portfolio stocks suggest that the shares we might like to short are indeed generally available, and eventually do attract substantial interest from short-sellers. However, we find that short interest comes with a lag, consistent with informational frictions facilitating the disconnect across markets. These results are presented in Table A9 and described in more detail in Appendix D.

3.4. Return decomposition

Given the failure of common risk factors, priced characteristics, and limits to arbitrage arguments to explain our main results, we instead turn to more affirmative tests of an information-based mechanism in which loan prices reflect material non-public information about the firm that equity markets ignore. Specifically, we ask, if loan prices lead equities in terms of information revelation, then what is the source of information being captured by loan market participants? Table 4 serves to decompose loans' predictive content into different categories of information yet to be released about the firm. Specifically, we add forward-looking controls for various information releases, and then record the extent to which these controls absorb the predictive content of loan returns. As each information release is added to the set of controls, we can then interpret the associated attenuation in the loading on loan returns as the portion of the predictability attributable to that information type.

To accomplish this, we re-estimate Table 3, this time adding the next 12 months' worth of the following value-relevant news as separate regression controls: 1) the number of analyst forecasts that are revised up, net of the number revised down in a given month, scaled by the total number of forecasts (for fiscal year end earnings forecasts), 2) the reporting of SUE relative to a seasonal random walk model (Livnat and Mendenhall, 2006)

in months with earnings news (we deem non-earnings months as having zero earnings surprises), and 3) changes to long-term Standard and Poor's issuer credit ratings. To keep the sample consistent across models, we require that firms have analyst coverage and are rated as of the date of the loan return (i.e., at the end of month t).

For purpose of comparison, column 1 in Table 4 replicates our baseline specification from Table 3 in a pooled panel regression with time (year $\times$ month) fixed effects.²² Consistent with our earlier findings, we find a large and significant coefficient on the syndicated loan return predictor. In specifications 2 to 6, we add twelve forward-looking controls for 2) the net percentage of analysts with upward earnings forecast revisions, 3) SUE, 4) SUE and net analyst forecast revisions combined, 5) the monthly change in numerical issuer credit ratings, and 6) all of the above.

Beginning with columns 2, 3, and 4, we find that controlling for future changes in analyst forecasts and unexpected earnings over the next year significantly reduces the predictive power of loan returns. In particular, we can absorb almost 43% of the predictive information content contained in loan returns by controlling for these two forward-looking information sources (column 4). The associated reduction in the loading on loan returns is significant at the 1% level.²³ The observed attenuation suggests that nearly half of loan investors' informational advantage over public equity markets is associated with relatively near term earnings – those captured in analysts' forecast revisions and actual earnings surprises announced over the next 12 months. This is consistent with the seasonality in Figure 3, which suggests much of the predictability is centered during months where earnings are announced to public equity investors.

In column 5, we add leading controls for credit rating downgrades. To the extent that price changes in the loan market capture information about future cash flows, they should most directly reflect the partition of cash flow news that impacts firms' capacity to repay

²²We opt to report pooled regressions in place of Fama-Macbeth specifications because several of our forward-looking controls lack variation within at least one calendar month, prohibiting estimation of parameters for those months.

²³To test significance, we jointly estimate the specification with the baseline in column 1 as a system of seemingly unrelated equations. We then test the null hypothesis that the difference in the loan return coefficients across equations is equal to zero.

outstanding debt. Meanwhile, equity market investors respond sharply to news about credit risk when it becomes available (see, for example, Holthausen and Leftwich (1986)). Column 5 shows that among the candidate controls considered, the predictive content of loan returns most strongly captures information about future debt rating downgrades over the next year – information that equity markets also anticipate, but with a lag relative to private debt holders. Including controls for the numerical change in credit ratings over the next 12 months reduces the independent predictive content of loan returns by over 56%. Hence, anticipation of ratings changes alone explains one-third more of the predictive content of loan returns than SUE and analyst forecasts combined. When all three controls are included together in column 6, the information is sufficient to almost completely account for the predictive content of loan returns.

We find similar results when we split the tests in Table 4 into negative and positive realizations of future analyst revisions, earnings surprises, and credit rating changes in Appendix Table A10. In particular, we demonstrate that the explanatory power of these measures primarily resides in negative future realizations, which lines up with the larger magnitude of alphas on the short side of the trading strategy. This finding is also consistent with the conclusions of Kothari, Shu, and Wysocki (2009), who show that managers frequently communicate positive news to public markets while holding back negative information. Furthermore, the relative importance of credit rating downgrades is consistent with the findings of Ederington and Goh (1998), who show that stock analysts revise their forecasts following credit rating changes, but that only downward revisions are followed by systematically lower realized earnings.

Finally, as a confirmation that loan returns are capturing forward-looking information, column 7 replaces leading variables with contemporaneous and lagged controls. In contrast to the forward-looking controls, these have no effect on the predictive power of loan returns. This rules out the unlikely explanation that period t loan returns and $t + 1$ equity returns both respond to stale public information with a delay, but that loan returns simply do so one month ahead of equities. Consistent with this conclusion, in Appendix Table A11 we find that in a pooled cross-sectional regression of loan returns

on the same leading and lagging measures, it is future realizations of analyst revisions, SUE, and ratings changes that exhibit a positive significant relation.²⁴

Taken together, the results in Table 4 suggest that secondary loan market prices incorporate value-relevant information about future earnings, analyst forecast revisions, and changes in credit ratings ahead of equity markets. Given the motivating background for the paper (i.e., lenders' insider status), it is not surprising that loan prices should reflect material non-public information in advance of announcement to public markets. However, what is more puzzling is the fact that equity investors fail to extract this information from loan prices and promptly incorporate it into equity valuations, as suggested by standard theory (e.g., Radner (1979), Grossman and Stiglitz (1980), Hellwig (1980)). Next, we explore some potential explanations.

4. What Facilitates Integration?

On average, our evidence suggests that equity markets ignore the price signal available from traded private debt, leading to a failure of integration across two important claims in a firm's capital structure. So what frictions prevent equity-holders from incorporating information in loan prices? Put otherwise, what types of settings might help facilitate market integration?

We begin by examining the hypothesis that attention may help facilitate market integration. Using media reporting of loan prices, as well as more general drivers of firm-specific attention, we test the extent to which making the firm or its loan prices more salient can induce equity markets to more carefully incorporate the information in loan prices. We continue by considering the hypothesis that, over and above attention, integration requires a form of expertise by testing for market integration across claims held

²⁴Specifically, we find that loan returns exhibit strong positive predictive relationships with future realizations of analyst revisions (1 month ahead), SUE (up to 3 quarters ahead), and credit rating up/downgrades (up to 3 quarters ahead). Moreover, we find that future realizations of SUE and credit rating changes are jointly positive, with large respective *F*-statistics and associated *p*-values of less than 0.01 in both cases. In contrast, we find almost no statistical evidence that lags of these variables over the preceding 12 months are positively related to loan returns (the only exception is 12 month lagged SUE).

by managers and funds that are simultaneously active in equity and loan markets. The evidence we present suggests the existence of a constraint on cross-market information processing that is reduced for funds that are active across both markets.

4.1. Attention

Investor inattention has been proposed as a central friction in investor information processing, potentially generating either over-reaction or under-reaction to public signals (Hirshleifer, Lim, and Teoh (2011)). The limited response to loan prices could reflect equity investors who are unaware of the secondary market for loans or, to the extent that they are aware, believe that little can be learned from paying attention to loan markets. Indeed, it is true that for the modal loan, daily and monthly returns are exactly zero. If tracking prices in this market imposes costs on equity traders, that may go a long way in explaining the delayed response we observe.

If the frictions preventing full and timely market integration are rooted in inattention, then when syndicated loan market information – particularly information about price movements – is made salient, we would expect markets to direct selective attention to the news and predictability to dissipate. To test this, we focus on weekly loan price movements reported in the Wall Street Journal. Using the same LSTA/Thomson Reuters loan market data we use in this paper, between August 2000 and August 2015, the Wall Street Journal published a weekly feature reporting the 25 biggest movers in the secondary loan market (“biggest movers” were ranked on absolute value change in the average bid reported by the LSTA). Because the timing used to construct the list is inconsistent (sometimes the ranking is done Monday through Friday, other times Tuesday to Tuesday) and because on occasion, loans that should have been on the list based on the reported methodology are excluded for unexplained reasons, we resort to transcribing the WSJ list by hand.

Table 5 replicates Table 2 using only the list of names reported in the biggest movers column for that month and hence focuses the analysis on names for which loan market prices would have been easily observable and more salient to equity market participants.

A few modifications to the strategy are necessary. First, we limit ourselves to two portfolios (winners and losers) based on whether or not the loan appreciated during the month. Second, in months for which we have less than three names in either portfolio, we instead invest the portfolio at the risk-free rate until the next month.

The returns to the long-short portfolio based on this basic strategy are large, earning monthly alphas between 2.078 and 2.564% across value- and equal-weighted strategies. If anything, we find that returns to the simpler newspaper strategy are larger than the returns to the full portfolio reported in Table 2.²⁵ If we believe that appearing in the WSJ serves as a meaningful shock to attention, or at least to the cost of paying attention, then the evidence here would seem inconsistent with inattention driving the delayed integration of news across markets.

In Table 6, we report the analogous result in the form of a Fama-Macbeth specification (column 1) and then move on to consider other sources of variation in attention to the firm. Perhaps selective attention to the firm, for example due to increased media coverage or other firm events, leads investors to direct more time to the analysis of all firm information, including prices of related claims like private debt.

To test this idea, we use data on the monthly number of news article mentions for each firm in our sample. We focus on coverage in the Wall Street Journal, New York Times, USA Today, and Washington Post, following the prior literature on firms' national newspaper coverage (e.g., Fang and Peress (2009), Niessner and So (2017)).²⁶ We aggregate the number of separate news articles in each of these outlets each month. We then include the log of 1 plus this measure in our main Fama-MacBeth specification, both in levels and interacted with the loan return predictor, to assess how news flow affects the predictability we document. In column 2 of Table 6, we find that neither the level nor the interaction coefficient are statistically distinguishable from zero. In particular, the insignificant interaction term suggests that contemporaneous attention due to news cov-

²⁵The second row and fourth row of Table 5 confirm this by examining the returns to the 1 and 5 portfolios in Table 2, excluding names reported in the WSJ. Of course, the WSJ list also represents the most extreme loan returns; the long-short returns partially reflect that.

²⁶Thanks to Marina Niessner and Eric So for generously sharing these data.

Syndicated Loans: Past Week's Biggest Movers

Syndicated loans are corporate loans that are bought or traded by a group of banks and/or institutional investors. Investment-grade loans are investment-grade or unrated loans priced at or below the London interbank offered rate (Libor) plus 150 basis points (or 1.5 percentage points). Leveraged loans are speculative-grade or unrated loans priced at or above Libor plus 151 basis points. Below are the biggest gainers and losers among widely-quoted syndicated loans in secondary trading in the week ended Friday among the 226 loans with five or more bids. All loans listed are B-term, or sold to institutional investors.

Name	Loan rating Moody's/S&P	Coupon/interest (Libor + basis pts)	Maturity	Average bid (pct. pts.)	Weekly chg (pct. pts.)
Alpha Natural Resources	B1/BB-	L+275	May 31, '20	88.95	-0.36
Arch Coal Inc	B1/B+	L+500	May 17, '18	89.63	-1.04
Burger King Corp	(P)B1/B+	L+350	Sept. 15, '21	99.53	0.32
Caesars Entertainment Inc	B2/B+	L+525	April 2, '21	94.00	-0.60
Caesars Entertainment Inc	B2/CCC+	L+600	Sept. 24, '20	95.00	-0.60
Cequel Communications Holdings	Ba2/BB	L+275	Feb. 15, '19	98.60	0.41
CityCenter	B2/BB-	L+325	Oct. 16, '20	99.23	0.50
Dell Computer Corp	Ba2/BB+	L+350	March 25, '20	99.75	0.40
Delta Airlines	Ba1/BBB-	L+275	April 15, '17	98.92	0.31
Getty Images Inc	B2/B	L+350	Oct. 14, '19	93.46	1.38
Go Daddy Group	Ba3/B	L+375	April 30, '21	99.08	0.67
International Lease Finance Corp	N.R./N.R.	L+275	Feb. 18, '21	99.38	0.66
J. Crew	B1/B	L+300	Feb. 27, '21	96.00	0.92
Kronos Worldwide	B1/B+	L+375	Feb. 17, '20	99.98	0.31
Regal Cinemas	Ba1/BB	L+250	Aug. 23, '17	98.93	0.73
Reynolds Group	B1/B+	L+300	Dec. 15, '18	99.25	0.55
Sears Holdings Corp	B1/B	L+450	June 30, '18	96.50	-0.50
Seaworld Parks and Entertainment Inc	Ba3/BB	L+225	May 10, '20	94.96	0.86
Southwire Co	Ba3/BB+	L+250	Dec. 20, '20	97.40	-1.21
SunGard Availability	Ba3/BB-	L+500	March 27, '19	91.70	-0.38
SuperValu	B1/B+	L+350	March 21, '19	97.60	-0.44
Univar NV	B3/B+	L+350	June 30, '17	99.23	0.33
Virgin Media Investment Hldgs(NTL)	Ba3/BB-	L+275	Feb. 6, '20	97.79	0.54
Walter Energy Inc	B3/B-	L+575	March 14, '18	88.96	0.68
Zebra Technologies	Ba2/BB+	L+375	Oct. 2, '21	100.48	0.56

Note: These are the averages of indicative bid prices provided by bank-loan traders and expressed as a percentage of the par or face value. All ratings are for specific loans and not for the company. These prices do not represent actual trades nor are they offers to trade; rather they are estimated values provided by dealers; N.R. Indicates that this issue is not rated

Source: LSTA/Thomson Reuters MTM Pricing

Figure 4

Wall Street Journal Biggest Movers. Between August 2000 and August 2015, the Wall Street Journal printed a weekly table of the 25 "Biggest Movers" in the syndicated loan market. This figure provides an example from October 2014.

erage has little effect on the degree of price integration. Furthermore, the syndicated loan return coefficient remains economically and statistically significant after adding controls for news attention, ruling out yet another omitted variable explanation.

We also consider analyst coverage and analyst upgrades and downgrades as drivers of firm attention. For upgrades and downgrades, we construct a Net Analyst Upgrades variable, computed as the number of analyst EPS forecasts that were revised up in a given month, net of the number revised down, scaled by the total number of forecasts. The analyst coverage measure is simply the number of analysts with earnings forecasts outstanding in a given month, and is set to zero for firms with no analyst coverage. In columns 3 and 4, we add these measures in both levels and interacted with the loan return

predictor. In both cases, we find that the interaction term coefficients are economically small and statistically insignificant. In contrast, the syndicated loan return remains an economically and statistically significant predictor of next-month equity returns. Taken together, these results fail to support the idea that shocks to attention facilitate the equity market's prompt incorporation of information captured in loan prices.

4.2. Cross-market expertise

Our analysis to this point suggests that shocks to the salience of firm-specific information, including broadcasting of loan prices that are likely to incorporate private information, is insufficient for integrating the loan and equity markets. However, if shining a light on value-relevant information is not enough, then what is required to integrate these two markets? Our second hypothesis is one of specialization of expertise, whereby equity and loan market investors have unique skill sets, or perhaps believe that their information is more specialized than it really is. For example, Vives and Yang (2017)) develop a model featuring agents whose sophistication dictates their ability to comprehend the price function and thus recover information implicit in prices. This limited sophistication can equivalently be interpreted as an investor's choice to not pay attention to some prices in the economy.

In practice, our strategy is simple and requires little expertise. However, if equity traders believe that understanding loan prices requires additional background, they may choose to ignore the information available. Note that this is still a form of inattention. However, in contrast with an inattention hypothesis whereby relevant information is easily interpreted but not salient, our cross-market specialization hypothesis suggests that information can be prominently reported on and will still be willfully ignored by participants who believe they lack the expertise to act on it.

To test this, we look to market participants who trade across markets and therefore would have the expertise and wherewithal to take advantage of news embedded in loan prices. Specifically, we focus on hybrid equity funds that actively trade in equities but also maintain exposure to the syndicated loan market. We identify these funds by looking to

CRSP mutual fund holdings data and searching holdings for assets identified as syndicated loans. Scanning through the Lipper classifications for these funds and reading their prospectuses, we find funds that are generally active, describe themselves as balanced or hybrid funds, and have a mandate to invest in loans, bonds, and equities. Hereafter, we refer to these funds as “integrated funds.” At any given point in time, roughly 75% of our equity cross-section will be owned by at least one integrated fund. Based on the fact that mutual fund holdings data tracking syndicated loans begins in 2010, we have a shorter sample, but still enough to tease out some cross-sectional implications.

With integrated funds identified, we retest our market integration hypothesis for equities owned by integrated funds in the month prior to changes in loan prices for the corresponding firms. In Table 7, we test for a difference in market integration across equities owned by integrated funds versus the rest of the sample by re-running the Fama-Macbeth regressions from Table 3. We interact the loan return predictor with a dummy variable equal to one if an integrated fund owned the corresponding stock in the prior month, and zero otherwise. The implicit hypothesis is that these integrated funds will be more likely to understand the relevance of loan prices to equity values and, because of their existing exposure to a given stock, be predisposed to pay attention and act on that information.

Columns 1 to 3 present a variety of specifications, and in each case, the predictability of loan returns is offset by the interaction term on the integrated funds dummy. Column 1 presents the most basic specification, while column 2 adds controls for size, book-to-market, and lagged six-month returns. Finally, in column 3, because we might think that stocks owned by integrated funds are likely to be different along several dimensions (including arbitrage constraints), we add interactions for the characteristics associated with limits to arbitrage discussed in Section 3.3. and presented in Table A8. While we cannot rule out that the dummy for integrated funds captures some other important characteristic of the stocks that also happens to predict the degree of market integration, it is encouraging to note that the economic and statistical magnitude of the interaction is largely unaffected by the inclusion of other plausibly important characteristics like size,

liquidity, and institutional ownership as interactions. Thus, even the most conservative interpretation of the result would suggest that knowledge of who owns equities is a useful predictor of market integration across firms' capital structure.²⁷ Meanwhile, in each case, the level effect on loan returns is positive, significant, and larger than the Fama-MacBeth coefficients reported in Table 3. This suggests that the strategy to trading on loan market integration survives even late in the sample, but only for equities that are not owned by hybrid cross-market participants.

The effects of cross-market expertise show up at the intensive margin as well. In column 5, we condition on the sample of shares held by integrated funds and then interact the loan return predictor with the number of syndicated loans held by the associated funds. This gives us variation in the scale of integrated funds' involvement in the secondary loan market. We find that stocks held by integrated funds holding a larger number of loans – funds that are perhaps more involved in the secondary loan market – also exhibit less predictability. In contrast, we do not find that the level of market integration is higher among equities that have a larger proportion of their float held by integrated funds.

Given the evidence that cross-market expertise seems to be important to learning from prices in other markets, it is natural to ask how specific is that expertise. Further, is it easily transferable across funds in the same family or with the same manager? Regarding specificity, we consider whether the cross-market expertise among investors that are active in both the loan and equity markets extends to funds that invest in equities and debt more generally. Could it be, for example, that predictability is muted by more general cross-market expertise in corporate debt, as opposed to more specific expertise in a particular type of debt, i.e., private loans?

We examine this idea by defining an alternative integrated fund indicator that measures whether a given stock is held by funds managing both corporate bonds and equities. If the knowledge required to integrate the loan and equity markets is of a relatively gen-

²⁷Future work may be able to further explore the mechanism by which integrated funds integrate markets by documenting their trading patterns around loan returns. However, given that funds in our sample predominantly report quarterly holdings data, our ability to separate trading around monthly loan returns from trading around the lagged equity returns is limited to a handful of funds.

eral form, then investors who participate in both the corporate bond and stock markets should also possess it. However, we find that the alternative integrated fund indicator is equal to one for over 98% of the stock-month observations in our sample. Further, the observed predictability holds over this 98%, suggesting that a set of skills and knowledge more specialized than a simple understanding of the links between debt and equity markets is at play in our integrated funds result.

We also consider the transferability of this expertise across funds, either by way of fund family linkages, or by way of fund managers' portable human capital. For example, if a fund possesses the requisite expertise to extract information from loan prices, then does the family of funds possess the same? Prior work in the literature on mutual funds suggests that funds in the same family often have access to a common set of research and analysis (Brown and Wu, 2016). Further, Guedj and Papastaikoudi (2005) and Gaspar, Massa, and Matos (2006) show that funds within a family strategically interact in ways that might facilitate the proliferation of cross-market expertise. Alternatively, these coordination channels may be offset by competition between managers within a fund complex (e.g., Kempf and Ruenzi (2007)).

To examine this, we ask if the integrated fund result extends to the level of the fund family by considering all equities owned by any fund in the same family as an integrated fund. Under this broader definition, we again find that almost all (over 97%) of the stocks in our sample would be considered to be held by "integrated" investors. Given that our results are not driven by the remaining 3% of the sample, it follows that the skill and knowledge necessary to integrate markets does not seem to permeate across funds within the same family. Instead, the knowledge appears to be confined to funds that are actively involved in both markets.

Alternatively, we might hypothesize that expertise to utilize loan prices is a portable trait specific to fund managers, as opposed to some combination of managerial skill and fund-specific conditions. We use data on fund manager identities to disentangle the manager- and fund-specific channels by constructing a variable that indicates whether a stock is held by an integrated fund manager in their non-integrated funds. We then

include this indicator and its interaction with the loan return predictor in column 6, along with the integrated fund indicator and its interaction with loan returns. In this horse-race, we continue to find that the level effect of loan returns is positive and significant. Furthermore, we find that the loan return interacted with the integrated fund indicator loads with a significantly negative coefficient. In contrast, we find that the interaction between the loan return and the integrated fund manager indicator is economically small and statistically indistinguishable from zero.

The absence of an interactive effect with the manager-specific indicator suggests that the cross-market expertise necessary to effectively utilize cross-market information is not an individual-specific characteristic that is portable across funds. This result is reminiscent of prior findings on the portability of knowledge workers' performance. For example, Groysberg, Lee, and Nanda (2008) and Groysberg (2010) find that star analysts exhibit a decline in performance after switching brokerage firms. Huckman and Pisano (2006) show that the patients of cardiac surgeons who operate at multiple hospitals have lower mortality rates when their procedure is performed at the hospital where their surgeon's recent procedure volume is the highest.

Taken together, our results suggest that a combination of human capital and organizational structure, captured by the co-location of equity and loan desks within a fund, facilitates information transfer across markets. In particular, our results are consistent with fund-level cross-market expertise being the result of interaction between equity and loan specialists on the same portfolio management team, at the same time. Absent the opportunity to interact with one another in real time within the fund, the individual managers do not appear to be able to replicate the expertise they exhibit when working as part of a team. Although speculative, one interpretation of this fact might be that the personal interaction of asset managers across markets that occurs at integrated funds makes the acquisition and processing of cross-market price information less costly or more salient for equity managers in a way that news coverage or other shocks to attention do not.

5. Conclusion

This paper documents the new and surprising fact that private debt prices lead equities by as much as a month. In contrast with existing studies that demonstrate relatively tight integration across markets, we focus on a setting where market participants have disjoint information sets. We can therefore closely examine the adeptness of equity markets at extracting insiders' private information from debt prices. Meanwhile, perhaps equally surprising is that neither the salience of firms nor that of individual loan returns seems to have any effect on the level of market integration. Instead, the presence of funds with expertise across markets appears to be more important.

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Table 1

Summary Statistics. This table reports summary statistics for several key variables. Panel A reports the statistics for both the full sample of matched loans and stocks and the overall CRSP universe during the sample period. Panel B focuses on the sample of firm-month observations in the long and short portfolios. Syndicated loan return is defined as the percentage change in price for a given loan over a given month. Size is calculated as the log of market capitalization. Book-to-market is calculated as year-end book equity plus balance sheet deferred taxes scaled by the year-end market value of equity, imposing a six-month lag in measured values. Market leverage is calculated as the ratio of balance sheet debt to market value of common equity. The S&P Debt Rating is the long-term issuer credit rating variable for firms with rated debt. It is converted to a number ranging from 1 (AAA) to 23 (CC). The Bid-Ask Spread is calculated as the difference between ask and bid prices expressed as a percentage of stock price. The sample period is from September 1998 to August 2015. ***, **, and * respectively indicate significance at the 1, 5, and 10 % levels.

Panel A: Full Samples

Variable	Matched Loans and Stocks					Overall CRSP Sample					Diff (Mean)
	Mean	Median	Std. Dev.	10th ptile	90th ptile	Mean	Median	Std. Dev.	10th ptile	90th ptile	
Synd Loan Return (t), %	-0.078	0.027	2.480	-1.195	1.105	0.985	0.133	18.769	-15.856	16.670	-0.311**
Excess Stock Return (t+1), %	0.674	0.547	15.647	-15.165	15.897	12.725	12.648	2.160	9.996	15.572	1.581***
Size (log mkt cap, \$1000s)	14.306	14.233	1.264	12.806	15.945	0.891	0.600	2.204	0.180	1.586	-0.130***
Book-to-market	0.761	0.502	1.199	0.163	1.418	1.077	0.267	7.774	0.009	1.845	0.196***
Leverage	1.272	0.597	2.779	0.134	2.839	12.221	12	3.405	8	17	2.616***
S&P Debt Rating	14.837	15	1.881	13	17	1.689	0.455	2.996	0.033	4.762	-1.233***
Bid-Ask Spread, %	0.456	0.100	1.011	0.025	1.233						

Panel B: Extreme Portfolios, Matched Loan and Stock Sample

Variable	Long Portfolio					Short Portfolio					Diff (Mean)
	Mean	Median	Std. Dev.	10th ptile	90th ptile	Mean	Median	Std. Dev.	10th ptile	90th ptile	
Synd Loan Return (t), %	1.732	0.746	2.812	0.073	4.533	-1.850	-0.602	3.255	-5.085	-0.094	3.582**
Excess Stock Return (t+1), %	1.283	0.575	16.658	-15.791	17.833	-0.269	-0.150	19.034	-19.032	16.260	1.553***
Size (log mkt cap, \$1000s)	14.255	14.196	1.391	12.622	16.022	14.267	14.218	1.247	12.747	15.886	-0.012
Book-to-market	0.930	0.567	1.472	0.170	1.760	0.837	0.553	1.215	0.166	1.694	0.093***
Leverage	1.615	0.733	3.595	0.143	3.583	1.788	0.720	3.904	0.148	4.163	-0.173*
S&P Debt Rating	15.052	15	2.087	13	17	15.042	15	1.854	13	17	0.010
Bid-Ask Spread, %	0.520	0.113	1.117	0.026	1.498	0.567	0.114	1.227	0.026	1.646	-0.047*

Table 2

Equal Weighted Portfolio Returns. This table reports performance estimates of a trading strategy that sorts stocks on matched non-zero loan returns into quintiles. We report the performance of six equal-weighted portfolios: (i) the “Short” portfolio contains the quintile of stocks with the lowest observed loan returns, (ii) the “Long” portfolio contains the quintile of stocks with the highest observed loan returns, (iii) the “Long–Short” portfolio, which captures the difference in returns of the Long and Short portfolios, and (iv)–(vi) portfolios 2–4, which contain the second through fourth quintiles, respectively, of stocks sorted on observed loan returns. We report the raw returns for each of the portfolios. We also report CAPM alphas, as well as 3, 6, and 9-factor alphas for each of the portfolios. The 3-factor model includes the excess market return (RMR), the value factor (HML), and the size factor (SMB). The 6-factor model adds the momentum factor (UMD) as well as the short- and long-term reversal factors (STR and LTR). Finally, the 9-factor model further includes the liquidity (LIQ) and betting against beta (BAB), and distress risk factors. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Equal-weighted strategy	Raw Return	CAPM Alpha	3-factor Alpha	6-factor Alpha	9-factor Alpha
1 (Short)	-0.535 (-0.75)	-1.289 (-3.26)	-1.527 (-3.59)	-1.259 (-3.71)	-1.256 (-3.45)
2	0.922 (1.77)	0.352 (0.91)	0.116 (0.39)	0.230 (0.78)	0.120 (0.40)
3	0.926 (1.75)	0.358 (1.06)	0.188 (0.62)	0.280 (0.88)	0.102 (0.40)
4	1.052 (1.88)	0.469 (1.54)	0.233 (0.81)	0.308 (1.05)	0.222 (0.77)
5 (Long)	1.580 (2.40)	0.964 (2.12)	0.628 (1.70)	0.889 (3.09)	0.790 (2.80)
Long - Short	2.115 (4.63)	2.253 (4.83)	2.155 (4.77)	2.148 (4.89)	2.046 (4.43)
N months	204	204	204	204	204

Table 2 (continued)

Equal-weighted double sort	Raw Return	CAPM Alpha	3-factor Alpha	6-factor Alpha	9-factor Alpha
High loan spread	3.072 (4.91)	3.214 (5.09)	3.094 (4.93)	3.052 (5.01)	2.892 (4.63)
Low loan spread	0.939 (1.88)	1.011 (2.04)	0.906 (1.92)	1.014 (2.07)	0.932 (1.77)
Difference	2.133 (2.83)	2.203 (2.98)	2.188 (2.86)	2.038 (2.69)	1.960 (2.43)

Value-weighted strategy	Raw Return	CAPM Alpha	3-factor Alpha	6-factor Alpha	9-factor Alpha
1 (Short)	-0.482 (-0.60)	-1.224 (-3.07)	-1.253 (-3.25)	-1.061 (-2.92)	-0.979 (-2.90)
5 (Long)	0.874 (1.50)	0.341 (1.01)	0.212 (0.62)	0.338 (1.05)	0.308 (0.87)
Long - Short	1.356 (2.78)	1.565 (3.31)	1.465 (3.23)	1.400 (3.26)	1.287 (2.87)

Table 3

Fama Macbeth Predictive Regressions. This table reports estimates from Fama and MacBeth (1973) regressions. We regress excess stock returns in month $t + 1$ on the following regressors observable at the end of month t : syndicated loan return, log market capitalization at the end of the previous month, book-to-market ratio, lagged stock return over the previous six months, stock return in month t , the ratio of short- and long-term debt to market capitalization, expected default probabilities based on S&P credit ratings, a credit rating up/downgrade indicator (± 1 for up/downgrades), standardized unexpected earnings (SUE) announced in months $t - 2$ through t , an indicator for whether a firm has analyst coverage, Fama-French 3-factor loadings calculated using daily returns during the month, institutional ownership, and turnover. We report the time series average of cross-sectional R^2 s. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Excess Stock Return ($t+1$)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Synd Loan Return (t)	0.468 (2.84)	0.444 (2.46)	0.407 (2.14)	0.494 (2.61)	0.570 (2.40)	0.470 (2.61)	0.402 (2.32)	0.409 (2.36)	0.464 (2.48)	0.431 (2.45)	0.387 (2.33)	0.522 (2.81)
Size		-0.245 (-1.75)	-0.265 (-1.80)	-0.225 (-1.73)	-0.228 (-1.45)	-0.262 (-1.88)	-0.247 (-1.77)	-0.282 (-1.95)	-0.122 (-0.79)	-0.236 (-1.68)	-0.252 (-1.82)	0.001 (0.00)
Book-to-market		-0.505 (-1.46)	-0.481 (-1.38)	-0.437 (-1.31)	-0.808 (-2.21)	-0.535 (-1.54)	-0.530 (-1.57)	-0.509 (-1.43)	-0.265 (-0.80)	-0.545 (-1.56)	-0.436 (-1.29)	-0.613 (-1.31)
Lagged 6mRet		0.281 (0.32)	0.183 (0.19)	0.164 (0.19)	0.378 (0.43)	0.304 (0.37)	0.216 (0.27)	0.327 (0.37)	0.314 (0.36)	0.258 (0.29)	0.402 (0.46)	0.250 (0.22)
Stock Return (t)			-0.007 (-0.54)									-0.009 (-0.54)
Market Leverage				-0.201 (-1.46)								-0.132 (-0.83)
Expected Default Prob.					0.016 (0.69)							0.008 (0.27)
Credit Rating Up/Downgrade						0.726 (0.82)						0.073 (0.09)
SUE							0.041 (1.59)					0.026 (0.83)
Analyst Coverage								0.718 (1.57)				0.262 (0.46)

Table 3 (continued)

Excess Stock Return (t+1)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta(\text{RMRF})$									-0.435 (-1.49)			-0.564 (-1.54)
$\beta(\text{SMB})$									-0.029 (-0.29)			0.110 (0.92)
$\beta(\text{HML})$									0.203 (1.63)			0.340 (1.94)
IO										-0.417 (-0.43)		-1.653 (-1.37)
Turnover											0.090 (0.75)	0.272 (1.80)
Synd Loan Return $\times$ Size												-0.413 (-2.74)
Constant	0.879 (1.72)	4.097 (1.98)	4.326 (1.99)	3.984 (2.14)	3.827 (1.63)	4.354 (2.14)	4.160 (2.03)	4.021 (1.90)	2.585 (1.12)	4.354 (2.05)	4.160 (2.00)	1.667 (0.55)
Avg R-squared	0.028	0.108	0.133	0.141	0.149	0.122	0.129	0.121	0.174	0.123	0.128	0.365
N obs	18,335	18,335	18,335	17,883	15,798	18,335	18,335	18,335	18,335	18,335	18,335	15,448
N months	204	204	204	204	204	204	204	204	204	204	204	204

Table 4

Return Decomposition. This table decomposes the predictability documented in earlier tables by including twelve monthly leading (and lagging) controls for various firm measures. Column 1 presents a pooled baseline regression of next-period equity returns on current period loan returns with year $\times$ month fixed effects. Columns 2 through 5 add twelve leading variables for the net % of analysts revising fiscal year forecasts up (net of those revising down) in a given month, the SUE relative to a seasonal random walk model in any earnings announcement month (set to zero in non-earnings months), net analyst revisions and SUE together, and the numerical change in long-term issuer credit rating in any given month. Column 6 includes all leading controls together. Column 7 repeats the exercise replacing leads with lags plus a contemporaneous version of the controls. All regressions include controls for size, value, and momentum as in the prior table, alongside year $\times$ month fixed effects. The t -statistics reported in parentheses below the coefficient estimates are computed using standard errors double clustered at the firm and year level. % attenuation refers to the percentage reduction in the coefficient on Synd Loan Return relative to Column 1. *** indicates the coefficient is significantly different from the baseline coefficient in (1) at the 1% level.

	(1) Baseline	(2) % Net analyst revisions [t+1, t+12]	(3) SUE [t+1, t+12]	(4) SUE and net revisions [t+1, t+12]	(5) Δ LT credit ratings [t+1, t+12]	(6) All controls [t+1, t+12]	(7) All controls [t-12, t]
Synd Loan Return (t)	0.275 (2.74)	0.211 (2.14)	0.186 (1.86)	0.157 (1.60)	0.119 (1.24)	0.062 (0.65)	0.277 (2.72)
% Attenuation		23.27%***	32.36%***	42.90%***	56.72%***	77.45%***	-0.72%
Observations	15,461	15,461	15,461	15,461	15,461	15,461	15,461
R-squared	0.214	0.256	0.231	0.265	0.242	0.281	0.218

Table 5

WSJ Sample. This table reports performance estimates of the long-short portfolio formed using only names reported in the Wall Street Journal's "Biggest Movers" column. In Panel A, the portfolio returns are value-weighted. In Panel B, the portfolio returns are equal-weighted. We report the raw returns for each of the portfolios. We also report CAPM alphas, as well as 3, 6, and 9-factor alphas for each of the portfolios. The 3-factor model includes the excess market return (RMRF), the value factor (HML), and the size factor (SMB). The 6-factor model further adds the momentum factor (UMD) as well as the short- and long-term reversal factors (STR and LTR). Finally, the 9-factor model further includes factors capturing liquidity (LIQ), betting against beta (BAB), and distress. The *t*-statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

EW Long-Short Portfolio Returns	Raw Return	CAPM Alpha	3-factor Alpha	6-factor Alpha	9-factor Alpha
WSJ List	2.334 (3.00)	2.404 (3.04)	2.249 (2.84)	2.384 (3.41)	2.119 (2.87)
Non-WSJ List	1.767 (3.70)	1.904 (4.18)	1.643 (4.61)	1.710 (4.60)	1.482 (3.99)
VW Long-Short Portfolio Returns	Raw Return	CAPM Alpha	3-factor Alpha	6-factor Alpha	9-factor Alpha
WSJ List	2.458 (3.16)	2.564 (3.32)	2.349 (3.45)	2.514 (4.30)	2.078 (3.23)
Non-WSJ List	1.315 (2.44)	1.451 (2.76)	1.226 (2.73)	1.222 (2.67)	1.045 (2.20)

Table 6

Attention Measures. This table reports estimates from Fama and MacBeth (1973) regressions. We regress excess stock returns in month $t + 1$ on the following regressors observable at the end of month t : syndicated loan return, log market capitalization at the end of the previous month, book-to-market ratio, and lagged stock return over the previous six months. We also include an indicator for whether the firm appeared in the WSJ Movers table during the month, the log of 1 plus the total number of news articles covering the firm across four national newspapers (USA Today, The Wall Street Journal, The New York Times, and The Washington Post) during the month, the number of positive earnings revisions (analyst upgrades) minus the number of negative earnings revisions (analyst downgrades) scaled by the number of analyst forecasts, and the number of analysts covering the firm. We report the time series average of cross-sectional R^2 s. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Excess Stock Return (t+1)	(1)	(2)	(3)	(4)
Synd Loan Return (t)	0.284 (2.04)	0.558 (2.74)	0.459 (2.78)	0.508 (2.11)
Size	-0.296 (-2.11)	-0.245 (-1.70)	-0.259 (-1.85)	-0.195 (-1.29)
Book-to-market	-0.235 (-0.81)	-0.509 (-1.52)	-0.603 (-1.71)	-0.486 (-1.33)
Lagged 6mRet	-0.388 (-0.45)	0.312 (0.39)	0.231 (0.28)	0.357 (0.42)
Synd Loan Return $\times$ WSJ Mover	0.107 (0.95)			
WSJ Mover	0.151 (0.23)			
Synd Loan Return $\times$ News Articles		0.013 (0.04)		
News Articles		-0.108 (-0.37)		
Synd Loan Return $\times$ Net Analyst Upgrades			-0.160 (-1.58)	
Net Analyst Upgrades			0.285 (1.13)	
Synd Loan Return $\times$ Number of Analysts				0.029 (0.55)
Number of Analysts				-0.040 (-1.10)
Constant	4.584 (2.25)	4.155 (1.93)	4.339 (2.11)	3.569 (1.60)
Avg R-squared	0.139	0.133	0.136	0.138
N obs	17,115	18,335	18,335	18,335
N months	181	204	204	204

Table 7

Integrated Funds. This table reports estimates from Fama and MacBeth (1973) regressions. We regress excess stock returns in month $t+1$ on the following regressors observable at the end of month t : syndicated loan return, an integrated mutual fund indicator, log market capitalization at the end of the previous month, book-to-market ratio, lagged stock return over the previous six months, idiosyncratic volatility, institutional ownership, bid-ask spread, and illiquidity. The integrated mutual fund indicator is equal to one if an integrated fund owned the corresponding stock in the prior month, and zero otherwise. Integrated funds are defined as funds holding both stocks and syndicated funds. The integrated fund manager indicator is equal to one if a stock is held by a non-integrated fund whose manager has also managed an integrated fund, and zero otherwise. We report the time series average of cross-sectional R^2 s. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Excess Stock Return (t+1)	(1)	(2)	(3)	(4)	(5)	(6)
Synd Loan Return (t)	1.738 (3.11)	1.417 (2.64)	2.235 (2.66)	0.729 (1.27)	-0.375 (-0.63)	1.705 (1.64)
Synd Loan Return $\times$ Integrated Fund	-1.840 (-3.11)	-1.436 (-2.40)	-2.461 (-2.58)			-2.634 (-1.95)
Synd Loan Return $\times$ Loans Held				-0.348 (-1.93)		
Synd Loan Return $\times$ Shares Held					0.210 (0.34)	
Synd Loan Return $\times$ Integrated Fund Manager						0.506 (0.40)
Integrated Fund	0.193 (0.40)	0.158 (0.29)	0.690 (1.26)			0.628 (1.24)
Loans Held				-0.125 (-1.16)		
Shares Held					0.016 (0.18)	
Integrated Fund Manager						-0.057 (-0.17)

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	(1)	(2)	(3)	(4)	(5)	(6)
Size		0.055 (0.29)	0.081 (0.33)	0.076 (0.31)	0.031 (0.13)	0.090 (0.40)
Book-to-market		-0.857 (-2.32)	-0.598 (-1.86)	-0.596 (-1.37)	-0.512 (-1.10)	-0.597 (-1.68)
Lagged 6mRet		0.368 (0.37)	0.336 (0.41)	1.190 (1.14)	0.954 (0.94)	0.379 (0.46)
IVOL			0.784 (2.74)	1.260 (3.76)	1.202 (3.39)	0.708 (2.10)
Log(IO)			-0.087 (-0.23)	-0.382 (-0.64)	-0.325 (-0.58)	-0.031 (-0.08)
Bid-Ask Spread			-0.250 (-1.17)	-0.303 (-1.46)	-0.309 (-1.40)	-0.225 (-1.04)
Illiquidity			-0.950 (-0.68)	0.019 (0.01)	1.953 (0.90)	-0.671 (-0.45)
Synd Loan Return $\times$ Size			-0.145 (-0.20)	-0.772 (-0.78)	-0.638 (-0.68)	-0.276 (-0.38)
Synd Loan Return $\times$ IVOL			3.832 (0.68)	8.813 (1.19)	5.319 (0.65)	5.342 (0.88)
Synd Loan Return $\times$ Log(IO)			1.027 (0.39)	4.276 (0.89)	1.897 (0.40)	0.362 (0.14)
Synd Loan Return $\times$ Bid-Ask Spread			-0.161 (-0.43)	-0.293 (-0.38)	-0.093 (-0.12)	-0.191 (-0.49)
Synd Loan Return $\times$ Illiquidity			0.109 (0.12)	2.104 (1.39)	2.116 (1.32)	0.292 (0.25)
Constant	1.182 (1.69)	0.730 (0.24)	0.067 (0.02)	1.239 (0.29)	1.633 (0.41)	0.142 (0.04)
Avg R-squared	0.061	0.132	0.267	0.315	0.314	0.288
N obs	5,633	5,633	5,631	4,432	4,432	5,631
N months	60	60	60	60	60	60

Internet Appendix:

Equity Price Discovery with Informed Private Debt

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A Weekly Portfolio Sort Results

In this appendix section, we re-run our portfolio sorts at a weekly horizon in order to better understand why our results differ from those of Allen and Gottesman (2006). We calculate weekly loan returns and sort all stocks with a matched loan return in week t into quintiles. We then form quintile equity portfolios as in our tests at the monthly frequency, and track their performance over the following week. Consistent with the findings of Allen and Gottesman (2006), we find that loan returns fail to predict abnormal returns in the equity market at a weekly frequency. In particular, we find that a long-short portfolio at the weekly frequency generates an insignificant 9-factor alpha of 0.130% (t -statistic = 0.95).

To reconcile this finding with our baseline results, we also calculate and plot cumulative abnormal returns at the weekly frequency. This analysis also helps us to understand how long it takes for information contained in loan returns to enter equity prices. We first calculate the average 9-factor alpha from a long-short portfolio formed using weekly loan returns as the predictor, as in Allen and Gottesman (2006), but implemented with a delay of 1 to 16 weeks (~ 4 months). We then calculate cumulative abnormal returns over each horizon. The results are plotted in Figure A1, and show that the information contained in loan returns in week t appears to be gradually incorporated into equity prices over the following 10 weeks. After this, abnormal long-short equity returns oscillate in a relatively random manner.

Taken together, these results indicate that the predictability we document at the monthly frequency coexists with the non-result of Allen and Gottesman (2006), and that the difference in findings is driven by a gradual diffusion of value-relevant information over a horizon exceeding one week.

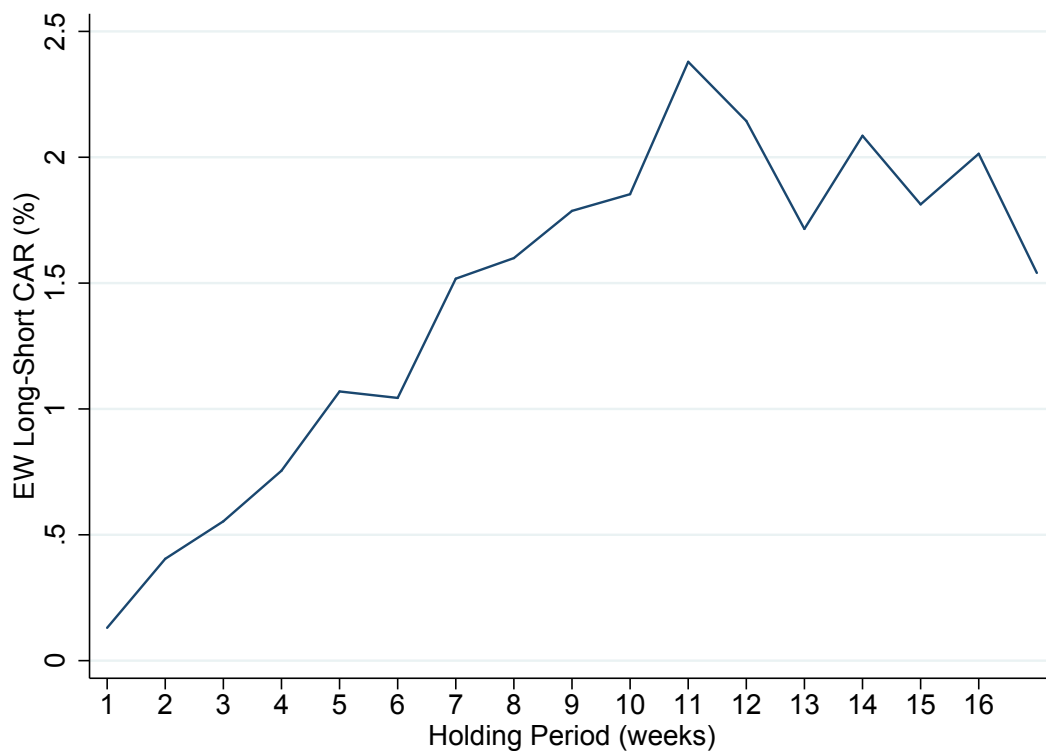


Figure A1

Cumulative Abnormal Returns, Weekly Strategy. We plot the cumulative abnormal returns from the equal weighted long-short portfolio over holding periods ranging from 1 to 16 weeks. We first calculate the average 9-factor alpha from a long-short portfolio formed using weekly loan returns as the predictor, but implemented with a delay of 1 to 16 weeks. We then calculate and plot cumulative abnormal returns over each horizon.

B Cross-Predictability of Loans and Equities

In this appendix section, we test the cross-predictability between loans and equities using a sample of loans with single dealer quotes. Importantly, most of the loan quotes we observe are averages of individual quotes submitted by multiple dealers. Thus, even in months with non-zero price movements, there may be some dealers who do not actually update their quotes. However, since we only use non-zero returns in our tests, prices based on a single dealer quote are likely to be non-stale.

In Table A3, we report the results from two Fama-MacBeth regressions for this subsample. For comparability across the columns, we standardize both equity returns and stock returns to have zero mean and unit standard deviation. In column 1, we replicate our main results on the restricted sample, regressing equity returns at $t+1$ on loan returns at t , plus lagged equity returns, size, and book-to-market. As in the full sample, loan returns lead equity returns. However, in column 2, where we regress $t+1$ loan returns on time t equity returns, loan returns, size, and book-to-market, we find that equities do not significantly predict subsequent loan returns. Moreover, from an economic perspective, the coefficient magnitude on loans in column 1 is much larger than that on stocks in column 2 (recall that equity and loan returns have been standardized to ensure comparability). In sum, while the sample is necessarily much smaller than in our baseline tests, controlling for loan market staleness, it does not appear that equities lead loans.

C Trading Strategy: Robustness Checks

In this appendix section, we present robustness checks for our trading strategy by restricting the sample of loans and stocks on several dimensions. In Table A4, row a, we show that the strategy performs well even for firms above the median NYSE size cutoff. In row b, we condition on loans containing a revolving credit facility. Revolvers might convey a natural advantage to loan participants, given that lenders might learn about draw-downs in advance. For example, revolvers secured with floating liens on collateral accounts (e.g., receivables) will provide lenders with a constant stream of information on the value of those accounts, information not available to equity markets (see Gustafson, Ivanov, and Meisenzahl (2016)). We find that trading on only the prices of revolving loan facilities generates slightly higher returns than the unconditional strategy. However, trading on the prices of term loans held by non-bank institutions that do not participate in revolvers still yields monthly alphas of 1.26 to 1.46% (row c).

In row d of Table A4, we restrict the trading strategy to equities that do not have earnings announcements in the holding period month. Although Figure 3 suggests a seasonal earnings component in the loan signals, we find that the strategy still earns substantial excess profits outside of firms' earnings announcement windows. Thus, while earnings news is an important component of the information in loan prices, it is not the entire story.

Loan prices might also provide early warnings about immediate distress. In some of the early important work using the secondary loan data, Gande, Altman, and Saunders (2006) show that, in the months leading up to firm defaults and bankruptcies, loan prices provide more timely indicators of the severity of distress. In row e of Table A4, we condition on borrowers outside of distress by dropping loans with prices below 90 (the threshold below which the Loan Syndications and Trading Association differentiates par from distressed loans) and again find a healthy result. This suggests that loan prices contain non-public information that is value-relevant even well outside of distress for levered firms, and that this information is not immediately recognized in equity prices.

To assess our trading strategy's potential for implementation over the sample period, we also consider alternative weighting schemes in forming the quintile portfolios. Specifically, in row f of Table A4, we calculate and report the average raw and risk-adjusted returns to the turnover-weighted long-short portfolio, where turnover is calculated over the month prior to portfolio formation. Since algorithmic traders may focus on stocks with high turnover to minimize the trading costs and price impact of their orders, we may find that the strategy returns are dampened when using this alternative weighting scheme. Instead, the estimates in row f indicate that the turnover-weighted trading strategy delivers statistically significant average alphas of over 2% per month.

Finally, it is natural to ask if the predictive content in loan prices we demonstrate is unique, or if it also appears in the prices of public bonds. To address this question, in the tests presented in Table A5, we merge our sample of firms (with both traded loans and equities) to those with bonds reported in the TRACE database. Note that merging with TRACE costs the sample in both the cross-section and the time series. In particular, TRACE data are only reported beginning in 2002, shortening our time series by nearly 25%. Among our remaining sample, only 42% of firm-month observations with non-zero loan returns have corresponding traded bonds.

We immediately note that among firms without traded bonds, the predictive power of loan returns on equities is robust. This result is presented in Table A5, column 1 in terms of portfolio sorts: the 9-factor alpha of the long-short portfolio formed using only firms without bonds is large, at 1.77% per month, and significant with a *t*-statistic of 3.49. As a first step, the presence of strong predictability among firms with no bonds outstanding affirms that at least some of the loan information generating predictability is unique to the private debt market. That is, the information contained in these loan returns cannot be reacting to information captured in public debt markets, since these firms have no bonds. Meanwhile, the returns to the strategy for firms with bonds are smaller, although the difference is not statistically significant. Given that the statistical and the economic significance of the predictability is concentrated among firms without bonds, it seems unlikely that controlling for bond returns will drive out the results from

Table 2.

Nevertheless, in column 1 of Table A6, we test this by re-examining the predictive effect of loans while controlling for any available bond market information. Using the baseline specification from Table 3 of the main paper, in column 1 we add information on bond returns as a control; for firms that have traded bonds, we use the actual return in month t and set the bond return to the cross-sectional mean for those without bonds.²⁸ We find that adding a control for bond market information has a limited effect on the predictability. Columns 2 and 3 demonstrate why: again, the predictive power of loan returns is largely concentrated among firms without bonds. This should not be surprising, given that our result is strongest among small to mid-size firms. Finally, in column 4 we focus on the subsample of firms with bonds. If the information embedded in loans was shared by bonds, we would expect to find bond returns leading equities when they are available. Instead, consistent with Kwan (1996) and Hotchkiss and Ronen (2002), we find no evidence that bonds lead equities.²⁹

²⁸We also add a dummy for whether or not bonds were outstanding such that our results are invariant to the number we impute for missing bond returns. We could equivalently set bond returns to the median or to zero when they are missing without changing the economic interpretation.

²⁹Note that, throughout the paper, we focus on non-zero loan returns, motivated by the fact that zero loan returns are more likely to be stale. However, in the merged bond sample, we allow for zero loan returns and zero bond returns. Excluding one would seem to require excluding both, which then limits the sample excessively.

D Risk Based Explanations & Limits to Arbitrage

In this appendix section, we explore alternative interpretations of the lead-lag relationship between loan and equity prices; notably risk and limits to arbitrage. Under the risk interpretation, changes in loan value might be associated with changing exposure to unobserved (and hence uncontrolled-for) risk factors. Under limits to arbitrage, predictability might be concentrated in shares of hard-to-trade stocks.

To test the risk based explanation, we examine the persistence of portfolio returns. If the abnormal performance of the long-short portfolio reflects mispricing that is eventually corrected, then the abnormal portfolio performance should exhibit a marked time decay when delaying portfolio formation. In contrast, the long-short portfolio return should exhibit a large degree of persistence if loan market returns signal changes in equityholders' exposure to systematic risks.

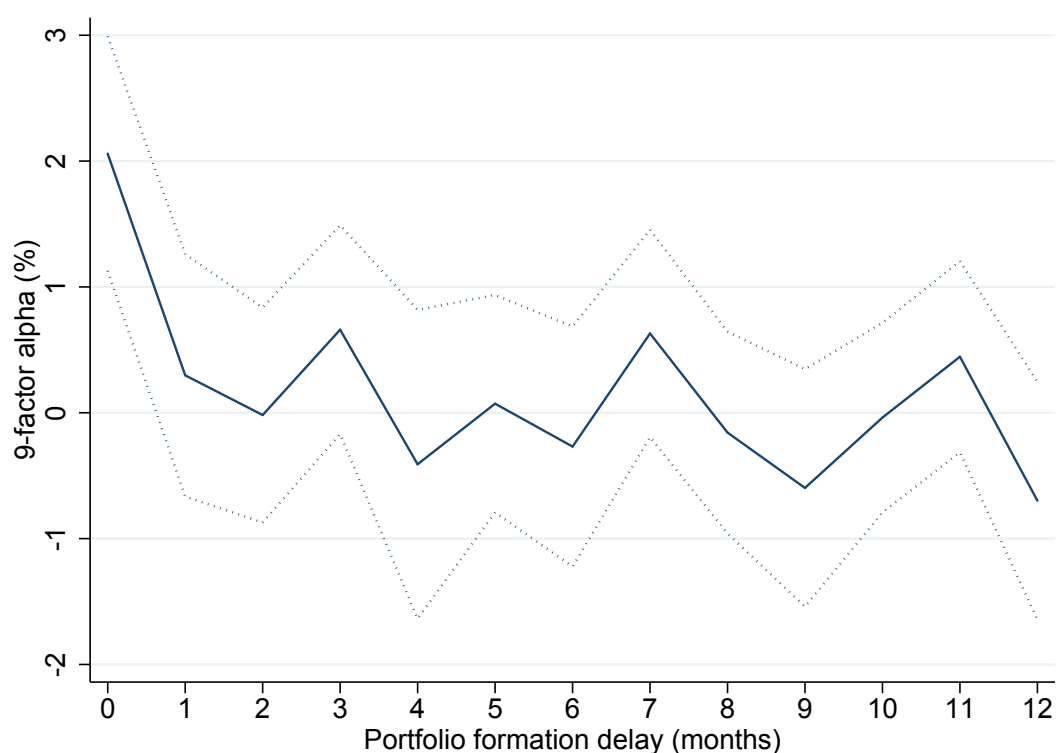


Figure A2

Delayed Portfolio Formation. The figure plots the 9-factor alpha from equal-weighted long-short portfolios formed based on the past month's loan returns (i.e., as in Table 2), alongside the returns from the same portfolio formed with a one to 12-month delay. Dashed lines surrounding the alpha estimates (solid line) represent the two standard error bands.

Figure A2 plots the effect of delaying the use of loan market signals observed in month t . The figure plots 9-factor alphas (solid line) as a function of the delay in portfolio formation (in months). We also show the two standard error bars (dashed lines) surrounding the alpha estimates. Consistent with the mispricing interpretation, we find that the strategy alphas exhibit strong time decay. Specifically, we find that even a one-month delay in portfolio formation, i.e., loan market signals in month t are used to form and hold portfolios during month $t + 2$ instead of $t + 1$, yields a long-short alpha that is statistically indistinguishable from zero. Similarly, further delays in portfolio formation yield long-short alphas that appear to randomly oscillate around zero.

Overall, given the lack of persistence in returns, our reading of Figure A2 seems less consistent with a risk based explanation and more consistent with equities failing to capture value-relevant information evident in loan prices. Further, it appears that the mispricing is corrected within about one month on average.

To test whether arbitrage constraints can help explain the relationship between loan returns and subsequent stock returns, we consider several standard proxies for characteristics associated with limits to arbitrage. The proxies are idiosyncratic volatility, institutional ownership, the bid-ask spread, and illiquidity. Following Campbell, Lettau, Malkiel, and Xu (2001), we calculate idiosyncratic volatility by fitting the three-factor Fama and French (1993) model using daily returns for each stock during month t . We calculate institutional ownership as the number of shares held by institutions in the Thomson Reuters 13F Holdings database at the end of the previous quarter divided by the total number of shares outstanding. The bid-ask spread is calculated as the difference between the Ask and Bid prices reported by CRSP as a percentage of the share price at the end of month t . Finally, following Amihud (2002), illiquidity is calculated as the average ratio of daily absolute return to dollar trading volume in the prior year. To reduce the distributional skewness of the institutional ownership measure, we take its natural log.³⁰

We include the interactions between each of the arbitrage constraint measures and

³⁰Our results are qualitatively unchanged whether or not we take this log transformation.

the loan return predictor in columns 1 through 4 of Table A8. As in Table 3, we demean the arbitrage constraint measures in each cross-section before computing the interactions. The syndicated loan return coefficients then measure the respective predictive effects for a firm with average idiosyncratic volatility, institutional ownership, bid-ask spread, and illiquidity. Across the four specifications, we find that the interactions between the arbitrage constraint proxies and the loan predictor are indistinguishable from zero at standard levels of significance. Further, the effect associated with loan returns continues to be positive and statistically significant in all cases, suggesting the predictive relation between the syndicated loan and stock markets is significant for firms with average arbitrage constraints.

In column 5, we simultaneously control for idiosyncratic volatility, institutional ownership, bid-ask spread, illiquidity, and size. Again, the estimates in column 5 indicate that simultaneously controlling for all the arbitrage constraint measures does not help explain the level effect associated with syndicated loan returns. For a hypothetical firm with average size and arbitrage constraints, the syndicated loan predictor remains an economically and statistically significant predictor of next-month excess stock returns (coefficient = 0.635; t -statistic = 2.42).

As an additional test of the effect of limits to arbitrage, we consider to what extent the cost of shorting shares of recent losers in the loan market serves as a barrier to implementation of the strategy. While the equal-weighted portfolio earns 0.81% to 1.58% on the long side, we find that the value-weighted strategy earns much of its returns from the short leg of the portfolio. The cost and availability of shares for shorting is therefore relevant to strategy profits. The dynamics of short interest in our portfolio, however, seem to suggest that the shares we might like to short in the portfolio are indeed generally available, and eventually do attract substantial interest from short-sellers. However, we find that interest comes with a lag, consistent with informational frictions facilitating the disconnect across markets.

To show this, Table A9 maintains the sample of loan-equity pairs from earlier tables, but replaces the dependent variable in Fama-Macbeth regressions with the change in the

equity short ratio over the month, and projects this on contemporaneous loan returns as well as lagged loan returns (column 1), alongside controls for size, value, momentum (column 2), and contemporaneous equity returns (column 3). The short ratio is calculated as the number of shares held short divided by total shares outstanding.³¹

In each column of Table A9, we see that short interest rises with poor loan returns, but not just in the current period. Holding current loan returns fixed, negative loan returns in the previous month are significantly associated with increases in short interest today with a magnitude nearly double that of the contemporaneous relationship. The fact that short interest responds to loan market information with a lag does not rule out the possibility that short selling costs reduce the profitability of the strategy, but it does suggest that, even facing those costs, short sellers find it profitable to respond to the news on these stocks eventually. Unless shorting costs happen to systematically decrease in the month following poor loan returns, the evidence would suggest that there is capacity to short poor performers in the loan market, but that short sellers react to the information content of syndicated loan returns with a lag.

³¹Changes in the short interest ratio are taken either at month end or (in the earlier part of the sample) between the 15th day of the month and the 15th day of the prior month. When changes in the short interest ratio are measured as of mid-month, loan returns (and lagged loan returns) are also measured as of mid-month.

Table A1

Equal Weighted Portfolio Factor Model Estimates. This table reports factor model estimates of a trading strategy that sorts stocks on matched non-zero loan returns into quintiles. We report the performance of three equal-weighted portfolios: (i) the “Short” portfolio contains the quintile of stocks with the lowest observed loan returns, (ii) the “Long” portfolio contains the quintile of stocks with the highest observed loan returns, and (iii) the “Long–Short” portfolio, which captures the difference in returns of the Long and Short portfolios. We report estimates for the CAPM as well as 3, 6, and 9-factor models for each of the portfolios. The 3-factor model includes the excess market return (RMRF), the value factor (HML), and the size factor (SMB). The 6-factor model adds the momentum factor (UMD) as well as the short- and long-term reversal factors (STR and LTR). Finally, the 9-factor model further includes factors capturing liquidity (LIQ), betting against beta (BAB), and distress. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Factor	Long (1)	Short (2)	Long-Short (3)	Long (4)	Short (5)	Long-Short (6)	Long (7)	Short (8)	Long-Short (9)	Long (10)	Short (11)	Long-Short (12)
Constant	0.964 (2.12)	-1.289 (-3.26)	2.253 (4.83)	0.628 (1.70)	-1.527 (-3.59)	2.155 (4.77)	0.889 (3.09)	-1.259 (-3.71)	2.148 (4.89)	0.790 (2.80)	-1.256 (-3.45)	2.046 (4.43)
RMRF	1.349 (10.40)	1.652 (13.43)	-0.302 (-2.23)	1.304 (14.60)	1.552 (14.40)	-0.248 (-2.20)	1.103 (12.58)	1.294 (12.51)	-0.192 (-1.80)	1.167 (13.12)	1.194 (11.80)	-0.027 (-0.24)
SMB				0.679 (4.56)	0.663 (4.42)	0.015 (0.11)	0.800 (7.71)	0.916 (8.12)	-0.117 (-0.75)	0.856 (5.86)	0.839 (6.13)	0.017 (0.09)
HML				0.702 (4.04)	0.283 (1.34)	0.419 (2.10)	0.565 (3.58)	0.225 (1.32)	0.340 (1.70)	0.439 (2.56)	0.166 (1.04)	0.272 (1.36)
UMD							-0.441 (-5.56)	-0.478 (-6.89)	0.037 (0.40)	-0.514 (-6.32)	-0.450 (-4.84)	-0.065 (-0.60)
ST REV							-0.047 (-0.60)	0.108 (0.76)	-0.155 (-1.19)	-0.037 (-0.43)	0.098 (0.66)	-0.135 (-0.96)
LT REV							-0.019 (-0.11)	-0.309 (-1.83)	0.290 (1.71)	0.028 (0.17)	-0.241 (-1.58)	0.269 (1.52)
LIQ										-0.005 (-0.09)	0.101 (1.55)	-0.106 (-1.50)
BAB										0.146 (1.66)	0.106 (0.66)	0.040 (0.31)
DISTRESS										0.048 (0.69)	-0.074 (-1.13)	0.122 (1.72)
Adj R-squared	0.581	0.608	0.048	0.697	0.656	0.093	0.777	0.738	0.105	0.779	0.741	0.114
N obs	204	204	204	204	204	204	204	204	204	204	204	204

Table A2

Value Weighted Portfolio Factor Model Estimates. This table reports factor model estimates of a trading strategy that sorts stocks on matched non-zero loan returns into quintiles. We report the performance of three value-weighted portfolios: (i) the “Short” portfolio contains the quintile of stocks with the lowest observed loan returns, (ii) the “Long” portfolio contains the quintile of stocks with the highest observed loan returns, and (iii) the “Long–Short” portfolio, which captures the difference in returns of the Long and Short portfolios. We report estimates for the CAPM as well as 3, 6, and 9-factor models for each of the portfolios. The 3-factor model includes the excess market return (RMRF), the value factor (HML), and the size factor (SMB). The 6-factor model adds the momentum factor (UMD) as well as the short- and long-term reversal factors (STR and LTR). Finally, the 9-factor model further includes factors capturing liquidity (LIQ), betting against beta (BAB), and distress. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Factor	Long (1)	Short (2)	Long-Short (3)	Long (4)	Short (5)	Long-Short (6)	Long (7)	Short (8)	Long-Short (9)	Long (10)	Short (11)	Long-Short (12)
Constant	0.341 (1.01)	-1.224 (-3.07)	1.565 (3.31)	0.212 (0.62)	-1.253 (-3.25)	1.465 (3.23)	0.338 (1.05)	-1.061 (-2.92)	1.400 (3.26)	0.308 (0.87)	-0.979 (-2.90)	1.287 (2.87)
RMRF	1.167 (9.16)	1.626 (12.78)	-0.459 (-2.99)	1.205 (13.24)	1.646 (13.55)	-0.441 (-3.54)	1.091 (10.45)	1.426 (12.20)	-0.335 (-2.86)	1.144 (13.31)	1.204 (11.03)	-0.060 (-0.53)
SMB				0.112 (0.80)	-0.007 (-0.03)	0.119 (0.57)	0.156 (0.93)	0.282 (1.74)	-0.126 (-0.59)	0.115 (0.53)	0.046 (0.23)	0.068 (0.26)
HML				0.447 (2.99)	0.136 (0.62)	0.311 (1.20)	0.360 (2.49)	0.168 (0.91)	0.191 (0.71)	0.298 (1.62)	0.133 (0.69)	0.165 (0.61)
UMD							-0.218 (-2.54)	-0.360 (-4.72)	0.142 (1.54)	-0.226 (-2.18)	-0.240 (-2.09)	0.013 (0.10)
ST REV							0.022 (0.28)	0.190 (1.28)	-0.168 (-1.06)	0.008 (0.10)	0.151 (1.01)	-0.143 (-0.83)
LT REV							0.029 (0.16)	-0.466 (-2.57)	0.495 (2.47)	0.046 (0.26)	-0.367 (-2.21)	0.413 (2.18)
LIQ										-0.150 (-2.27)	0.098 (1.03)	-0.248 (-2.11)
BAB										0.139 (0.93)	0.162 (1.01)	-0.024 (-0.12)
DISTRESS										-0.043 (-0.53)	-0.222 (-3.20)	0.180 (1.84)
Adj R-squared	0.514	0.580	0.069	0.547	0.578	0.077	0.565	0.641	0.108	0.577	0.651	0.148
N obs	204	204	204	204	204	204	204	204	204	204	204	204

Table A3

Cross-Predictability. This table reports the results from Fama-MacBeth regressions for a subsample of loans with single dealer quotes. In column 1, we regress equity returns at $t + 1$ on loan returns at t , plus lagged equity returns, size, and book-to-market. In column 2, we regress $t + 1$ loan returns on time t equity returns, loan returns, size, and book-to-market. For comparability across the columns, we standardize both equity returns and stock returns to have zero mean and unit standard deviation.

	Excess Stock Return (t+1) (1)	Excess Loan Return (t+1) (2)
Synd Loan Return (t)	0.181 (2.07)	-0.022 (-0.54)
Stock Return (t)	0.002 (0.07)	0.028 (1.60)
Size	0.002 (0.08)	-0.012 (-1.09)
Book-to-market	0.013 (0.44)	0.014 (0.58)
Avg. R-squared	0.266	0.339
N obs	5,001	5,001
N months	203	203

Table A4

Trading Strategy Sample Robustness. This table reports performance estimates, as described in Table 2, of various modifications of the equal weighted trading strategy. These include focusing on firms above the median NYSE size cutoff, limiting the loan signal to revolvers only, limiting the loan signal to institutional tranches of term loans only, trading only in equities without earnings announcements during the holding period, dropping distressed borrowers based on loan prices less than 90, and weighting portfolio constituents by equity turnover.

	EW Long-Short Portfolio Returns	Raw Return	CAPM Alpha	3-factor Alpha	6-factor Alpha	9-factor Alpha
(a) Above median size cutoff		0.872 (2.01)	0.894 (1.99)	0.832 (1.96)	0.851 (2.00)	0.773 (1.80)
(b) Revolvers		2.559 (2.93)	2.527 (2.79)	2.297 (2.89)	2.308 (2.88)	2.168 (2.96)
(c) Institutional Term Loan Bs		1.382 (3.52)	1.460 (3.76)	1.257 (2.59)	1.313 (3.00)	1.310 (2.78)
(d) Non-earning announcement months		1.962 (3.65)	2.147 (4.17)	1.933 (3.75)	1.985 (3.85)	1.997 (3.58)
(e) Non-Distressed Loans		1.783 (4.28)	1.861 (4.38)	1.765 (3.49)	1.806 (3.78)	1.738 (3.43)
(f) Turnover Weighted		2.273 (3.35)	2.520 (3.55)	2.092 (3.09)	2.177 (3.19)	1.924 (2.66)

Table A5

Portfolio Sorts: Bond Issuers vs. Non-Bond Issuers. This table reports double-sorted equal weighted portfolios formed on the basis of loan returns and then split across firms with and without traded bonds reported in the TRACE database.

	Long-Short (Not Bond Borrowers) (1)	Long-Short (Bond Borrowers) (2)	Difference (1)-(2)
Constant	1.776 (3.49)	0.623 (1.26)	1.153 (1.60)
RMRF	-0.194 (-0.92)	0.063 (0.37)	-0.258 (-0.97)
SMB	0.200 (0.63)	-0.043 (-0.14)	0.242 (0.44)
HML	-0.206 (-0.61)	0.192 (0.73)	-0.398 (-0.98)
UMD	0.103 (0.49)	-0.198 (-1.11)	0.301 (0.94)
ST REV	0.006 (0.03)	-0.507 (-3.15)	0.513 (1.75)
LT REV	0.111 (0.30)	0.087 (0.46)	0.024 (0.05)
LIQ	-0.040 (-0.28)	-0.254 (-2.86)	0.213 (1.52)
BAB	-0.346 (-2.14)	0.340 (1.58)	-0.687 (-2.41)
DISTRESS	-0.004 (-0.03)	-0.014 (-0.12)	0.009 (0.05)
Adj R-squared	0.003	0.084	0.055
N obs	158	158	158

Table A6

Fama-MacBeth Regressions: Bond Issuers vs. Non-Bond Issuers. We report Fama-Macbeth cross-sectional regressions of next period equity returns on current loan returns and bond returns, when available. When no bond return is available, we set the bond return equal to the mean for the time period and include a dummy for firms with bonds outstanding. Columns 2-4 compare the predictive power of loan returns across subsamples of firms with and without bonds outstanding. Distinct from other tables, here we include non-zero loan returns in the sample (as well as non-zero bond returns). We report the time series average of cross-sectional R^2 s. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Excess Stock Return (t+1)	(1)	No Bond Sample (2)	Bond Sample (3)	(4)
Synd Loan Return (t)	0.278 (2.05)	0.687 (2.19)	0.110 (0.73)	0.122 (0.77)
Bond Return (t)	0.060 (1.19)			0.060 (1.18)
Bond O/S	-0.020 (-0.14)			
Size	-0.155 (-1.91)	-0.046 (-0.31)	-0.187 (-2.24)	-0.200 (-2.50)
Book-to-market	-0.209 (-0.84)	0.072 (0.26)	-0.460 (-1.52)	-0.419 (-1.30)
Lagged 6mRet	-0.303 (-0.31)	-0.603 (-0.59)	0.057 (0.05)	-0.047 (-0.04)
Constant	0.863 (1.68)	0.721 (1.11)	0.575 (0.96)	0.779 (1.45)
Avg. R-squared	0.083	0.095	0.098	0.116
N obs	26,992	10,757	16,410	16,308
N months	158	158	158	158

Table A7

Panel Regressions. This table reports estimates from panel regressions with time and firm fixed effects. We regress excess stock returns in month $t + 1$ on syndicated loan returns in month t . Specification 1 includes no control variables. Specifications 2-5 include controls for log market capitalization at the end of the previous month, book-to-market ratio, and lagged stock return over the previous six months. Other control variables included in each specification match those in the table and column identified in the Control Specification row. Coefficients for these additional control variables are suppressed for brevity. In specification 5, the level effects for both syndicated loan returns and integrated funds include time interactions to control for time variation in the predictive effect of syndicated loan returns and isolate the cross-sectional effect of integrated investors. The interaction term between loan returns and the dummy for integrated funds captures the level shift in the predictive content of loan returns which is related to integrated funds, after conditioning out, among other things, the time series trends in predictability for stocks not held by integrated funds. The t -statistics reported in parentheses below the coefficient estimates are computed using standard errors that are clustered by both firm and time.

Excess Stock Return (t+1)	(1)	(2)	(3)	(4)	(5)
Synd Loan Return (t)	0.232 (2.45)	0.208 (2.24)	0.279 (3.06)	0.255 (3.37)	
Synd Loan Return $\times$ Integrated Fund					-1.347 (-2.46)
Integrated Fund					
Size		-3.168 (-6.68)	-2.595 (-4.76)	-3.068 (-6.23)	-3.792 (-4.81)
Book-to-market		0.513 (1.45)	0.119 (0.19)	0.453 (1.27)	0.483 (0.57)
Lagged 6mRet		-1.542 (-2.97)	-0.782 (-1.60)	-1.559 (-3.13)	-4.293 (-4.28)
Control Specification	Table 3, Col 1	Table 3, Col 2	Table 3, Col 12	Table A8, Col 5	Table 7, Col 3
R-squared	0.261	0.270	0.280	0.277	0.329
N obs	18,335	18,335	15,448	17,980	5,631
N months	204	204	204	204	60

Table A8

Arbitrage Constraints. This table reports estimates from Fama and MacBeth (1973) regressions. We regress excess stock returns in month $t+1$ on the following regressors observable at the end of month t : syndicated loan return, idiosyncratic volatility, institutional ownership, bid-ask spread, illiquidity, log market capitalization at the end of the previous month, book-to-market ratio, and lagged stock return over the previous six months. Idiosyncratic volatility (IVOL) is calculated by fitting the three-factor Fama and French (1993) model using daily returns for each stock during month t . Institutional ownership in month t is calculated as the number of shares held by institutions in the Thomson Reuters 13F Holdings database at the end of the previous quarter divided by the total number of shares outstanding. Bid-ask spread is calculated as the difference between the Ask and Bid prices reported by CRSP as a percentage of share price at the end of month t . Illiquidity is calculated as the average ratio of daily absolute return to dollar trading volume in the prior year (Amihud, 2002). We report the time series average of cross-sectional R^2 s. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Excess Stock Return (t+1)	(1)	(2)	(3)	(4)	(5)
Synd Loan Return (t)	0.377 (2.03)	0.630 (2.78)	0.465 (2.09)	0.783 (3.03)	0.635 (2.42)
Size	-0.286 (-1.90)	-0.256 (-1.83)	-0.281 (-2.14)	-0.342 (-2.43)	-0.261 (-1.64)
Book-to-market	-0.440 (-1.25)	-0.500 (-1.41)	-0.819 (-2.09)	-0.633 (-1.72)	-0.824 (-2.00)
Lagged 6mRet	0.537 (0.62)	0.248 (0.30)	0.125 (0.16)	0.246 (0.28)	0.357 (0.47)
IVOL	-0.271 (-2.22)				-0.209 (-1.42)
Log(IO)		-1.210 (-1.04)			-0.945 (-0.71)
Bid-Ask Spread			-0.306 (-0.34)		-0.308 (-0.28)
Illiquidity				-0.146 (-0.86)	-0.315 (-1.76)
Synd Loan Return $\times$ IVOL	-0.047 (-0.29)				-0.197 (-0.95)
Synd Loan Return $\times$ Log(IO)		-0.733 (-0.73)			-4.063 (-1.52)
Synd Loan Return $\times$ Bid-Ask Spread			0.333 (0.32)		2.271 (1.28)
Synd Loan Return $\times$ Illiquidity				0.152 (0.57)	-0.244 (-0.59)
Synd Loan Return $\times$ Size					-0.096 (-0.49)
Constant	5.206 (2.36)	5.090 (2.37)	4.859 (2.67)	5.576 (2.80)	5.675 (2.24)
Avg R-squared	0.151	0.142	0.153	0.138	0.268
N obs	18,329	18,335	17,986	18,335	17,980
N months	204	204	204	204	204

Table A9

Short Interest. This table reports estimates from Fama and MacBeth (1973) regressions of changes in equity short-interest, scaled by shares outstanding on loan returns in the same period, loan returns from the prior period, and controls for size, value, momentum, and contemporaneous equity returns. Changes in the short interest ratio are taken either at month end or (in the earlier part of the sample) between the 15th day of the month and the 15th day of the prior month. When changes in the short interest ratio are measured as of mid-month, loan returns (and lagged loan returns) are also measured as of mid-month. We report the time series average of cross-sectional R^2 s. The t -statistics reported in parentheses below the coefficient estimates are computed using Newey and West (1987) adjusted standard errors using a six-month lag.

Δ Short Interest Ratio (t)	(1)	(2)	(3)
Synd Loan Return (t)	-0.036 (-1.72)	-0.033 (-1.44)	-0.042 (-1.63)
Synd Loan Return (t-1)	-0.061 (-2.17)	-0.086 (-2.74)	-0.074 (-2.32)
Size		-0.023 (-1.60)	-0.020 (-1.32)
Book-to-market		0.071 (2.01)	0.065 (1.74)
Lagged 6mRet		-0.272 (-3.44)	-0.261 (-3.45)
Stock Return (t)			0.000 (0.12)
Avg R-squared	0.057	0.133	0.161
N obs	11,991	11,991	11,991
N months	203	203	203

Table A10

Return Decomposition, Signed Realizations. This table decomposes the predictability by including twelve monthly leading controls for various firm measures, as in Table 4, but using signed realizations. Column 1 presents a pooled baseline regression of next-period equity returns on current period loan returns with year $\times$ month fixed effects. Column 2 (3) adds twelve leading realizations of net % of analysts revising fiscal year forecasts down (up) in a given month. Column 4 (5) includes negative (positive) SUE relative to a seasonal random walk model in any earnings announcement month (set to zero in non-earnings months). Column 6 (7) includes the numerical change in long-term issuer credit rating for credit rating downgrades (upgrades) in any given month. Column 8 includes all signed leading controls together. All regressions include controls for size, value, and momentum as in the prior table, alongside year $\times$ month fixed effects. The t -statistics reported in parentheses below the coefficient estimates are computed using standard errors double clustered at the firm and year level. % attenuation refers to the percentage reduction in the coefficient on Synd Loan Return relative to Column 1. *** indicates the coefficient is significantly different from the baseline coefficient in (1) at the 1% level.

	(1) Baseline	(2) % Net analyst revisions (down) [t+1, t+12]	(3) % Net analyst revisions (up) [t+1, t+12]	(4) SUE (down) [t+1, t+12]	(5) SUE (up) [t+1, t+12]	(6) Δ LT credit ratings (down) [t+1, t+12]	(7) Δ LT credit ratings (up) [t+1, t+12]	(8) All controls [t+1, t+12]
Synd Loan Return (t)	0.275 (2.74)	0.217 (2.21)	0.240 (2.41)	0.185 (1.87)	0.273 (2.72)	0.099 (1.03)	0.274 (2.74)	0.046 (0.50)
% Attenuation		21.09%***	12.73%***	32.73%***	0.73%	64.00%***	0.36%	83.27%***
Observations	15,461	15,461	15,461	15,461	15,461	15,461	15,461	15,461
R-squared	0.214	0.249	0.240	0.231	0.218	0.244	0.218	0.290

Table A11

Loan Return Decomposition. This table presents results from a regression of loan returns (t) on twelve monthly leading and lagging, as well as contemporaneous, controls for various firm measures. All coefficients are from a single pooled regression with year $\times$ month fixed effects and controls for size, value, and momentum. Leading and lagging firm measures include the net % of analysts revising fiscal year forecasts up (net of those revising down) in a given month, the SUE relative to a seasonal random walk model in any earnings announcement month (set to zero in non-earnings months), and the numerical change in long-term issuer credit rating in any given month. The t -statistics reported in parentheses below the coefficient estimates are computed using standard errors double clustered at the firm and year level. F -statistics with p -values in brackets are from a Wald test of coefficients in a column being jointly equal to zero.

Lead/lag (months)	% Net analyst revisions [t , $t+12$]	% Net analyst revisions [$t-1$, $t-12$]	SUE [t , $t+12$]	SUE [$t-1$, $t-12$]	Δ LT credit ratings [t , $t+12$]	Δ LT credit ratings [$t-1$, $t-12$]
0	0.066 (1.20)		0.022 (1.79)		0.438 (2.55)	
1	0.087 (2.10)	-0.005 (-0.11)	0.004 (0.41)	-0.013 (-1.41)	0.393 (3.32)	-0.107 (-0.71)
2	-0.018 (-0.36)	0.053 (1.21)	0.011 (1.06)	-0.010 (-0.82)	0.228 (2.22)	-0.170 (-1.27)
3	0.012 (0.29)	-0.012 (-0.24)	-0.004 (-0.50)	0.004 (0.54)	0.035 (0.34)	-0.258 (-2.16)
4	0.053 (1.33)	0.023 (0.65)	0.016 (2.00)	0.004 (0.47)	0.134 (1.67)	-0.159 (-2.29)
5	-0.018 (-0.56)	0.001 (0.01)	0.008 (1.14)	0.012 (1.31)	0.088 (0.94)	-0.040 (-0.41)
6	-0.025 (-0.59)	0.009 (0.23)	0.013 (1.91)	0.008 (0.68)	0.179 (2.14)	-0.168 (-2.11)
7	-0.050 (-1.09)	-0.014 (-0.29)	0.010 (1.55)	-0.001 (-0.09)	0.103 (1.69)	-0.023 (-0.25)
8	-0.018 (-0.38)	0.032 (0.81)	0.013 (2.00)	-0.002 (-0.27)	0.301 (3.75)	-0.056 (-0.76)
9	-0.031 (-0.76)	0.074 (1.48)	0.010 (1.79)	-0.013 (-1.68)	0.010 (0.16)	-0.127 (-2.17)
10	-0.018 (-0.43)	-0.001 (-0.05)	-0.000 (-0.05)	-0.009 (-1.35)	0.171 (2.54)	-0.122 (-1.87)
11	0.015 (0.35)	0.001 (0.01)	-0.013 (-1.95)	-0.001 (-0.11)	0.066 (1.14)	-0.027 (-0.35)
12	0.028 (0.76)	-0.121 (-2.17)	-0.005 (-0.58)	0.014 (2.59)	0.092 (1.52)	-0.033 (-0.38)
F -statistic	0.75	1.02	2.43	1.02	4.26	3.31
p -value	[0.7118]	[0.4341]	[0.0045]	[0.4335]	[0.0000]	[0.0002]
Controls	Yes					
Observations	15,461					
R-squared	0.357					